

ON THE UNIFORM POSITIVITY OF F -SIGNATURE UNDER REDUCTION MODULO p

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Dedicated to Professor Karen E. Smith on the occasion of her sixtieth birthday.

ABSTRACT. Carvajal-Rojas, Schwede and Tucker asked whether the mod p reductions of a complex klt type singularity have uniformly positive F -signature for all but finitely many primes p . In this paper, we give an affirmative answer to this conjecture for pure subrings of regular local rings, including reductive quotient singularities. We also show that the conjecture can be reduced to the Gorenstein case. Finally, we discuss the connection with F -alpha invariants for log Fano pairs, introduced by Pande as characteristic p analogs of Tian's alpha invariants.

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1. INTRODUCTION

F -signature is an important numerical invariant of singularities in positive characteristic. Suppose that $(A, \mathfrak{m}_A)$ is a d -dimensional F -finite Noetherian local domain of characteristic $p > 0$ with perfect residue field $A/\mathfrak{m}_A$. Recall that an $\mathbb{F}_p$ -algebra B is said to be F -finite if the Frobenius endomorphism $F: B \rightarrow B$ is finite. This is a mild finiteness condition that is automatically satisfied for rings essentially of finite type over a perfect field. We fix an algebraic closure $\overline{\text{Frac}(A)}$ of the fractional field $\text{Frac}(A)$ of A , and for each integer $e \geq 1$, consider the extension ring $A^{1/p^e} = \{x \in \overline{\text{Frac}(A)} \mid x^{p^e} \in A\}$ of A . Let a_e denote the largest number of free copies

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of A appearing as direct summands of A^{1/p^e} . Then the limit $\lim_{e \rightarrow \infty} a_e/p^{de}$ is known to exist by Tucker [Tuc12] and is called the F -signature $s(A)$ of A . A key feature of F -signature is a result of Aberbach–Leuschke [AL03], which states that $s(A) > 0$ if and only if A is strongly F -regular, a fundamental class of singularities in positive characteristic that is viewed as a characteristic p counterpart of klt type singularities. In this paper, we use the method of reduction modulo p to define F -signature in equal characteristic zero and investigate its properties.

For the purposes of this introduction, consider the following ring:

$$R = (\mathbb{C}[X_1, \dots, X_n]/(f_1, \dots, f_r))_{(X_1, \dots, X_n)},$$

where each $f_i \in \mathbb{Z}[X_1, \dots, X_n]$ is a polynomial with integer coefficients. The reduction of R modulo a prime p , denoted by $R \bmod p$, is obtained by reducing the coefficients of the f_i modulo p , that is,

$$R \bmod p = (\mathbb{F}_p[X_1, \dots, X_n]/(f_1 \bmod p, \dots, f_r \bmod p))_{(X_1, \dots, X_n)},$$

where $f_i \bmod p$ denotes the image of f_i under the natural map $\mathbb{Z}[X_1, \dots, X_n] \rightarrow \mathbb{F}_p[X_1, \dots, X_n]$. The F -signature $s(R)$ of R is then defined as

$$s(R) = \liminf_{p \rightarrow \infty} s(R \bmod p).$$

Since it is not known whether the limit exists except in certain specific cases (see [Sin05], [CSTZ24], [BCPT25]), we take the limit inferior. More generally, for an arbitrary local ring R essentially of finite type over $\mathbb{C}$, the notion of reduction modulo p is defined in Definition 2.10, and the F -signature $s(R)$ is defined in Definition 4.1. However, in the introduction, we restrict ourselves to the above concrete special case for simplicity of exposition.

It is natural to ask how this invariant relates to klt type singularities, because such singularities conjecturally correspond to strongly F -regular singularities via reduction modulo p :

Conjecture 1.1 ([SS10, Question 7.2]). *Spec R is of klt type if and only if $R \bmod p$ is strongly F -regular for all but finitely many p .*

The “only if” direction is known to hold by [Tak04b]. The “if” direction also holds if R is $\mathbb{Q}$ -Gorenstein (see [HW02], [Smi97]), or more generally, if the anti-canonical ring of R is Noetherian (see [CEMS18]); however, the general case remains open even in dimension three.

In light of this (partly conjectural) correspondence and the result of Aberbach–Leuschke mentioned above, Carvajal-Rojas, Schwede and Tucker [CRST18] proposed the following conjecture:

Conjecture 1.2 ([CRST18, Question 5.9]). *If Spec R is of klt type, then $s(R) > 0$.*

By the “only if” direction of Conjecture 1.1, if Spec R is of klt type, then $R \bmod p$ is strongly F -regular, hence $s(R \bmod p) > 0$. Nevertheless, the limit of $s(R \bmod p)$ as $p \rightarrow \infty$ may be zero, which presents a serious obstacle. In dimension two, Conjecture 1.2 is straightforward, as two-dimensional klt type singularities are finite quotient

singularities, for which the F -signature is explicitly computable. Singh [Sin05] proved Conjecture 1.2 when R is a localization of a toric ring, Caminata, Shideler, Tucker and Zeman [CSTZ24] proved it for diagonal hypersurfaces, and Brosowsky, Coskun, Pande and Tucker [BCPT25] proved it for certain binomial hypersurfaces. However, very little is known beyond these cases. In this paper, we further investigate Conjecture 1.2.

Before stating the main result of this paper, we recall the definition of pure ring extensions. A ring extension $R \hookrightarrow S$ is said to be *pure* if the natural map $M \rightarrow M \otimes_R S$ is injective for all R -modules M . The main result of this paper is the following:

Theorem 1.3 (Theorem 4.7). *Let $(R, \mathfrak{m}) \hookrightarrow (S, \mathfrak{n})$ be a pure local $\mathbb{C}$ -algebra homomorphism between normal local rings essentially of finite type over $\mathbb{C}$. If $\text{Spec } S$ is of klt type and $s(S) > 0$, then $s(R) > 0$.*

Theorem 1.3 leads to the following applications concerning Conjecture 1.2:

Corollary 1.4 (Corollaries 4.9, 4.16). (1) *Conjecture 1.2 holds when R is a pure subring of a regular local ring essentially of finite type over $\mathbb{C}$ (e.g., a reductive quotient singularity).*
 (2) *Conjecture 1.2 can be reduced to the case where R is Gorenstein.*

We give a sketch of the proof of Theorem 1.3 below. The first key ingredient is the notion of local F -alpha invariants, a local version of F -alpha invariants introduced by Pande [Pan23], which are characteristic p analogs of Tian's alpha invariants. For the ring R considered above, the local F -alpha invariant $\alpha_F(R)$ is defined as

$$\alpha_F(R) = \liminf_{p \rightarrow \infty} \inf_{f \in \mathfrak{m} \bmod p} \text{fpt}(R \bmod p, f) \overline{\text{ord}}_{\mathfrak{m} \bmod p}(f),$$

where f runs through all nonzero elements in the maximal ideal $\mathfrak{m} \bmod p$ of the local ring $R \bmod p$. Here, $\text{fpt}(R \bmod p, f)$ denotes the F -pure threshold of f , a positive characteristic analog of the log canonical threshold, and $\overline{\text{ord}}_{\mathfrak{m} \bmod p}(f)$ denotes the normalized order of f with respect to $\mathfrak{m} \bmod p$ (see Definition 3.1 for its definition). Following Pande's argument, one can show that $s(R) > 0$ if and only if $\alpha_F(R) > 0$. Thus, the problem is reduced to proving the positivity of the local F -alpha invariant.

The second key ingredient is the notion of ultra- F -regularity, originally introduced by Schoutens [Sch05], which is a type of singularity over $\mathbb{C}$ defined in terms of the purity of ultra-Frobenii. An *ultra-Frobenius* is a ring homomorphism from R constructed as the ultraproduct of Frobenius maps in characteristic p over all primes p . Ultra- F -regularity has two main advantages. First, it descends under pure morphisms, whereas purity itself is not generally preserved under reduction modulo p . Second, in our setting, ultra- F -regularity enables us to verify the strong F -regularity of the pairs $(R \bmod p, f_p^{t_p})$, where $f_p \in \mathfrak{m} \bmod p$ is a nonzero element and $t_p \geq 0$ is a real number, simultaneously for almost all p (see Proposition 4.13). Here, the phrase “almost all p ” is made precise by fixing a non-principal ultrafilter on the set of all primes (see § 2.3 for details). Since the F -pure threshold $\text{fpt}(R \bmod p, f_p)$ is the supremum of the real numbers $t_p \geq 0$ such that the pair $(R \bmod p, f_p^{t_p})$ is strongly

F -regular, one can use ultra- F -regularity to show that

$$\mathrm{fpt}(R \bmod p, f_p) \geq \mathrm{fpt}(S \bmod p, f_p)$$

for all but finitely many p . This is a crucial step in the proof of Theorem 1.3.

We also provide a geometric interpretation of Conjecture 1.2. Schwede–Smith [SS10] introduced a global analog of strongly F -regular pairs, called *globally F -regular pairs*, and showed that if (X, Δ) is a log Fano pair over $\mathbb{C}$, then its reduction $(X \bmod p, \Delta \bmod p)$ modulo p is globally F -regular for all but finitely many p . Pande’s F -alpha invariant for a log Fano pair (X, Δ) , denoted by

$$\alpha_F((X, \Delta); -(K_X + \Delta))$$

in this paper, is formulated in terms of *global F -split thresholds* of the reductions $(X \bmod p, \Delta \bmod p)$. Given a globally F -regular pair (V, B) and a $\mathbb{Q}$ -Weil divisor D on V , the global F -split threshold $\mathrm{gfst}((V, B); D)$ of D with respect to (V, B) is defined as

$$\mathrm{gfst}((V, B); D) := \sup\{t \geq 0 \mid (V, B + tD) \text{ is globally } F\text{-regular}\}.$$

By considering Kollár components, we have the following theorem:

Theorem 1.5 (Theorem 5.2). *If $\alpha_F((X, \Delta); -(K_X + \Delta)) > 0$ for all projective log Fano pairs (X, Δ) over $\mathbb{C}$, then Conjecture 1.2 holds.*

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2. PRELIMINARIES

2.1. Birational geometry. In this subsection, we briefly review some basic terminology from birational geometry.

Let X be a normal variety over a field k of characteristic zero. Given a $\mathbb{Q}$ -Weil divisor D on X , the graded ring $\bigoplus_{i \geq 0} H^0(X, \mathcal{O}_X([iD]))$ is called the *section ring* of D and is denoted by $R(X, D)$.

Definition 2.1. (i) Let Δ be an effective $\mathbb{Q}$ -Weil divisor on X such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. Take a log resolution $\pi : Y \rightarrow X$ of the pair (X, Δ) . Then (X, Δ) is said to be *klt* if $\mathrm{ord}_E(K_Y - \pi^*(K_X + \Delta)) > -1$ for all prime divisors E on Y . This definition is independent of the choice of the log resolution π .

- (ii) X is said to be of *klt type* if there exists an effective $\mathbb{Q}$ -Weil divisor Δ on X such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier and (X, Δ) is klt.
- (iii) X is said to have only *klt singularities* if $-K_X$ is $\mathbb{Q}$ -Cartier and $(X, 0)$ is klt.

Definition 2.2. Suppose that X is projective over k .

- (i) Let Δ be an effective $\mathbb{Q}$ -Weil divisor on X such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. The pair (X, Δ) is said to be *log Fano* if $-(K_X + \Delta)$ is ample and (X, Δ) is klt.
- (ii) X is said to be of *Fano type* if there exists an effective $\mathbb{Q}$ -Weil divisor Δ on X such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier and (X, Δ) is log Fano.
- (iii) We say that X is *$\mathbb{Q}$ -Fano* if $-K_X$ is ample $\mathbb{Q}$ -Cartier and X has only klt singularities.

2.2. F -singularities. In this subsection, we recall basic notions concerning F -singularities. The reader is referred to [TW18] and [SS10] for a general overview.

Given a module M over a ring R of prime characteristic p , let $F_*^e M$ denote the R -module corresponding to the pushforward $F_*^e \widetilde{M}$ of the quasi-coherent sheaf $\widetilde{M}$ on $X := \operatorname{Spec} R$ corresponding to M via the e -times iterated Frobenius morphism $F^e : X \rightarrow X$. Moreover, if R is a Noetherian normal domain and D is a Weil divisor on $X = \operatorname{Spec} R$, then the R -module $H^0(X, \mathcal{O}_X(D))$ is denoted by $R(D)$.

First we recall the definition of test ideals and strongly F -regular singularities.

Definition 2.3 (cf. [HY03], [Tak04b]). Suppose that R is a Noetherian normal domain of characteristic $p > 0$ and Δ is an effective $\mathbb{Q}$ -Weil divisor on $\operatorname{Spec} R$. We assume that R is *F -finite*, that is, the Frobenius map $F : R \rightarrow R$ is finite. Let $\mathfrak{a}$ be a nonzero ideal of R and $t \geq 0$ be a real number.

- (i) For every R -module M , the submodule $0_M^{*(\Delta, \mathfrak{a}^t)}$ of M consists of all elements $z \in M$ for which there exists a nonzero element $c \in R$ such that for all large integers e , we have

$$F_*^e(c\mathfrak{a}^{\lceil tp^e \rceil}) \otimes z = 0 \in F_*^e R(\lceil p^e \Delta \rceil) \otimes_R M.$$

- (ii) The *test ideal* $\tau(R, \Delta, \mathfrak{a}^t)$ associated to the triple $(R, \Delta, \mathfrak{a}^t)$ is defined as

$$\tau(R, \Delta, \mathfrak{a}^t) = \bigcap_M \operatorname{Ann}_R 0_M^{*(\Delta, \mathfrak{a}^t)},$$

where M runs through all R -modules. When $\Delta = 0$, the ideal $\tau(R, \Delta, \mathfrak{a}^t)$ is denoted by $\tau(R, \mathfrak{a}^t)$. Similarly, when $\mathfrak{a} = R$, the ideal $\tau(R, \Delta, \mathfrak{a}^t)$ is denoted by $\tau(R, \Delta)$. If both $\Delta = 0$ and $\mathfrak{a} = R$, then $\tau(R, \Delta, \mathfrak{a}^t)$ is simply denoted by $\tau(R)$.

- (iii) The triple $(R, \Delta, \mathfrak{a}^t)$ is said to be *strongly F -regular* if $\tau(R, \Delta, \mathfrak{a}^t) = R$. The strong F -regularity of (R, Δ) , $(R, \mathfrak{a}^t)$ and R is defined similarly. Throughout this paper, all strongly F -regular rings are assumed to be Noetherian and F -finite.

Remark 2.4. If $(R, \mathfrak{m})$ is a local ring, then $\tau(R, \Delta, \mathfrak{a}^t) = \operatorname{Ann}_R 0_E^{*(\Delta, \mathfrak{a}^t)}$, where E is the injective hull of the residue field $R/\mathfrak{m}$ (see [BSTZ10, Definition-Proposition 3.3]; cf.

[HH90, Proposition 8.23]). The formation of test ideals commutes with localization and completion (see Remark 3.6 in loc. cit.; cf. [HT04, Propositions 3.1 and 3.2]).

Definition 2.5 ([Sch08b], [TW04]). Let R be a Noetherian F -finite integral domain of characteristic $p > 0$, f be a nonzero element of R and $t \geq 0$ be a real number.

- (1) We say that the pair $(R, (f)^t)$ is *sharply F -pure* if there exists an integer $e \geq 1$ such that the composition

$$R \rightarrow F_*^e R \xrightarrow{\cdot F_*^e f^{\lceil t(p^e-1) \rceil}} F_*^e R$$

of the e -times iterated Frobenius map $R \rightarrow F_*^e R$ and the multiplication by $F_*^e f^{\lceil t(p^e-1) \rceil}$ splits as an R -module homomorphism. When $f = 1$, we simply say that R is F -pure.

- (2) Assume that f is not a unit. Then the *F -pure threshold* $\text{fpt}(f)$ of f is defined by

$$\text{fpt}(f) = \sup\{t \in \mathbb{R}_{\geq 0} \mid (R, (f)^t) \text{ is sharply } F\text{-pure}\}.$$

If R is not F -pure, then we define $\text{fpt}(f) = 0$ for every f . When it is necessary to specify the base ring, we write $\text{fpt}(R, f)$ for $\text{fpt}(f)$.

Next, we introduce a global version of strong F -regularity and F -pure thresholds.

Definition 2.6. Let X be a Noetherian F -finite normal integral scheme of characteristic $p > 0$ and Δ be an effective $\mathbb{Q}$ -Weil divisor on X .

- (i) ([SS10]) The pair (X, Δ) is said to be *globally F -regular* if for every effective Weil divisor D on X , there exists an integer $e \geq 1$ such that the composite map

$$\mathcal{O}_X \rightarrow F_*^e \mathcal{O}_X \hookrightarrow F_*^e \mathcal{O}_X(\lceil (p^e - 1)\Delta + D \rceil)$$

splits as an $\mathcal{O}_X$ -module homomorphism, where the first map is the e -times iterated Frobenius map F^e and the second is the pushforward of the natural inclusion $\mathcal{O}_X \hookrightarrow \mathcal{O}_X(\lceil (p^e - 1)\Delta + D \rceil)$ by F^e .

- (ii) Suppose that (X, Δ) is globally F -regular. The *global F -split threshold* $\text{gfst}((X, \Delta); D)$ of an effective $\mathbb{Q}$ -Weil divisor D on X with respect to the pair (X, Δ) is defined as

$$\text{gfst}((X, \Delta); D) = \sup\{t \in \mathbb{R}_{\geq 0} \mid (X, \Delta + tD) \text{ is globally } F\text{-regular}\}.$$

When $\Delta = 0$, we simply write $\text{gfst}(X, D)$. If (X, Δ) is not globally F -regular, then we set $\text{gfst}((X, \Delta); D) = 0$ for any D .

Remark 2.7. F -pure threshold can be defined for pairs (X, D) , where X is a projective variety over an F -finite field and D is a Cartier divisor on X . However, the global F -split threshold is, in general, strictly smaller than the F -pure threshold. For instance, if k is a perfect field of characteristic two, $X = \mathbb{P}_k^3$, and $D \subseteq X$ is the hypersurface defined by $x^3 + y^3 + z^3 + w^3 = 0$, then $\text{fpt}(X, D) = 1$ since D is smooth, while $\text{gfst}(X, D) = 1/2$.

We now turn to the definition of the F -signature, the central object of study in this paper.

Definition 2.8 ([HL02] cf. [SVdB97]). Let $(R, \mathfrak{m}, k)$ be a Noetherian F -finite local ring of characteristic $p > 0$. For each integer $e \geq 1$, let a_e denote the largest number of free copies of R appearing as direct summands of $F_*^e R$, that is, $F_*^e R \cong R^{a_e} \oplus M_e$ where M_e is an R -module with no free direct summands. Then F -signature $s(R)$ of R is defined as

$$s(R) = \lim_{e \rightarrow \infty} \frac{a_e}{p^{e(d + [k:k^p])}}.$$

This limit is known to exist by [Tuc12].

Proposition 2.9 ([AL03]). *Let the notation be the same as in Definition 2.8.*

- (1) $0 \leq s(R) \leq 1$.
- (2) $s(R) = 1$ if and only if R is regular.
- (3) $s(R) > 0$ if and only if R is strongly F -regular.

In order to define F -signature in equal characteristic zero in Section 4, we recall the method of reduction modulo p . We briefly outline the construction below.

Definition 2.10. Let $x \in X$ be a (not necessarily closed) point of a variety X over $\mathbb{C}$. We say that $(A, x_A \in X_A)$ is a *model* of $x \in X$ if the following conditions hold:

- (i) A is a finitely generated $\mathbb{Z}$ -subalgebra of $\mathbb{C}$.
- (ii) X_A is a scheme of finite type over A and x_A is a point of X_A such that there exists a morphism $X \rightarrow X_A$ sending x to x_A making the following diagram commute:

$$\begin{array}{ccccc} \overline{\{x\}} & \hookrightarrow & X & \longrightarrow & \operatorname{Spec} \mathbb{C} \\ \downarrow & & \downarrow & & \downarrow \\ \overline{\{x_A\}} & \hookrightarrow & X_A & \longrightarrow & \operatorname{Spec} A, \end{array}$$

where $\overline{\{x\}}$ (resp. $\overline{\{x_A\}}$) denotes the closure of $\{x\}$ in X (resp. of $\{x_A\}$ in X_A) with the reduced induced scheme structure.

By generic freeness, after replacing A by a localization $A[1/a]$ at some element $a \in A$, we may assume that X_A and $\overline{\{x_A\}}$ are flat over A . For every closed point $\mu \in \operatorname{Spec} A$, we define X_μ as the scheme-theoretic fiber of $X_A \rightarrow \operatorname{Spec} A$ over μ , that is, $X_\mu = X_A \times_{\operatorname{Spec} A} \operatorname{Spec} A/\mu$. We refer to X_μ as a *reduction modulo p* of X . For a general closed point μ , the fiber $\overline{\{x_A\}} \times_{\operatorname{Spec} A} \operatorname{Spec} A/\mu$ of $\overline{\{x_A\}} \rightarrow \operatorname{Spec} A$ over μ is an integral closed subscheme of X_μ , and we define x_μ as its generic point. When X is affine and $(R, \mathfrak{m}) = \mathcal{O}_{X,x}$, we define $R_A := \mathcal{O}_{X_A}$ and $R_\mu := \mathcal{O}_{X_\mu, x_\mu}$, and write $\mathfrak{m}_A$ for the prime ideal of R_A corresponding to x_A . We refer to $(A, R_A, \mathfrak{m}_A)$ as a *model* of R .

If X is normal, then X_μ is a normal integral scheme for general closed points $\mu \in \operatorname{Spec} A$ by [Gro66, Théorème 12.2.4]. Then, for a $\mathbb{Q}$ -Weil divisor Δ on X , an argument similar to the one above yields a reduction Δ_μ of Δ , which is a $\mathbb{Q}$ -Weil divisor on X_μ . The reader is referred to [Gro66, §§11–12] and [HH20] for basic properties of reduction modulo p .

There is a significant correspondence between F -singularities and singularities in complex birational geometry via reduction modulo p .

Theorem 2.11. *Let X be a normal variety over $\mathbb{C}$ and Δ be an effective $\mathbb{Q}$ -Weil divisor on X such that $K_X + \Delta$ is $\mathbb{Q}$ -Cartier. Fix a model (A, X_A, Δ_A) of (X, Δ) .*

- (1) ([HW02], [Tak04b]) *(X, Δ) is klt if and only if (X_μ, Δ_μ) is strongly F -regular for general closed points $\mu \in \text{Spec } A$.*
- (2) ([SS10]) *Suppose that X is projective. If (X, Δ) is a log Fano pair, then (X_μ, Δ_μ) is globally F -regular for general closed points $\mu \in \text{Spec } A$.*

We will use Lemma 2.14 later in the proof of one of the main results of this paper. Although this lemma may be well known to experts, we include a proof here due to the lack of a suitable reference. Before stating it, we recall the definition of parameter test submodules and F -rational singularities.

Definition 2.12 (cf. [ST08]). Suppose that R is an F -finite integral domain with canonical module ω_R . Let $\mathfrak{a} \subseteq R$ be a nonzero ideal and $t \geq 0$ be a real number.

- (i) The *parameter test submodule* $\tau(\omega_R, \mathfrak{a}^t)$ of the pair $(R, \mathfrak{a}^t)$ is defined as the unique smallest nonzero submodule M of ω_R satisfying that $\text{Tr}_{F^e}(F_*^e \mathfrak{a}^{\lceil t(p^e-1) \rceil} M) \subseteq M$ for every integer $e \geq 1$, where $\text{Tr}_{F^e} : F_*^e \omega_R \rightarrow \omega_R$ is the trace map of the e -times iterated Frobenius map F^e .
- (ii) The pair $(R, \mathfrak{a}^t)$ is said to be *F -rational* if R is Cohen–Macaulay and $\tau(\omega_R, \mathfrak{a}^t) = \omega_R$.

Remark 2.13. As observed in [ST08, Proposition 6.5 (1)], if $(R, \mathfrak{a}^t)$ is strongly F -regular, then it is F -rational. Moreover, when R is Gorenstein, the converse also holds.

Lemma 2.14. *Let $R = \bigoplus_{n \in \mathbb{Z}} R_n$ be an F -finite $\mathbb{Z}$ -graded domain with unique homogeneous maximal ideal $\mathfrak{m}$. For every nonzero homogeneous ideal $\mathfrak{a}$ of R and every real number $t > 0$, the pair $(R, \mathfrak{a}^t)$ is F -rational if and only if so is the localization $(R_{\mathfrak{m}}, (\mathfrak{a}R_{\mathfrak{m}})^t)$ at $\mathfrak{m}$.*

Proof. Let A be an F -finite Cohen–Macaulay local (resp. $*$ local) domain A with canonical (resp. $*$ canonical) module ω_A , $I \subseteq A$ be a nonzero ideal and $s > 0$ be a real number. The parameter test ideal $\tau_{\text{par}}(A, I^s)$ is defined as

$$\tau_{\text{par}}(A, I^s) = \tau(\omega_A, I^s) :_A \omega_A \subseteq A,$$

where $\tau(\omega_A, I^s)$ is the parameter test submodule of the pair (A, I^s) . Note that $\tau_{\text{par}}(A, I^s) = A$ if and only if (A, I^s) is F -rational.

It is well known that R is Cohen–Macaulay if and only if so is $R_{\mathfrak{m}}$. Therefore, we may assume that R is Cohen–Macaulay. Pick a homogeneous test element $c \in R$. Then, by an argument similar to the proof of [HT04, Lemma 2.1], $\tau(\omega_R, \mathfrak{a}^t)$ can be written as

$$\tau(\omega_R, \mathfrak{a}^t) = \sum_{e \in \mathbb{N}} \sum_{\varphi^{(e)}} \varphi^{(e)}(F_*^e(c \mathfrak{a}^{\lceil tp^e \rceil} \omega_R)) \subseteq \omega_R$$

where $\varphi^{(e)}$ runs through all elements of $\text{Hom}_R(F_*^e \omega_R, \omega_R)$. Considering the homogeneous decomposition of $\varphi^{(e)}$, we may assume that each $\varphi^{(e)}$ is a homogeneous element. This description shows $\tau(\omega_R, \mathfrak{a}^t)$ is a homogeneous ideal of R , and its formation commutes with localization. The same holds for $\tau_{\text{par}}(R, \mathfrak{a}^t)$. Thus, $(R, \mathfrak{a}^t)$

is not F -rational if and only if $\tau_{\text{par}}(R, \mathfrak{a}^t) \subseteq \mathfrak{m}$, which is equivalent to saying that $\tau_{\text{par}}(R_{\mathfrak{m}}, (\mathfrak{a}R_{\mathfrak{m}})^t) \subseteq \mathfrak{m}R_{\mathfrak{m}}$. This last condition means that the localized pair $(R_{\mathfrak{m}}, (\mathfrak{a}R_{\mathfrak{m}})^t)$ is not F -rational. $\square$

2.3. Ultraproducts. In this subsection, we quickly review basic notions on ultraproducts and their applications to commutative algebra. We refer the reader to [Sch03], [Sch08a] and [Yam25a, Section 3] for details. Throughout this subsection, let W be an infinite set, R be a local ring essentially of finite type over $\mathbb{C}$ and $\mathcal{P}$ denote the set of all prime numbers.

Definition 2.15. A non-empty subset $\mathcal{F}$ of the power set of W is said to be a *non-principal ultrafilter* on W if $\mathcal{F}$ satisfies the following four conditions:

- (i) If $A \in \mathcal{F}$ and $A \subseteq B \subseteq W$, then $B \in \mathcal{F}$.
- (ii) If $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$.
- (iii) If A is a finite subset of W , then $A \notin \mathcal{F}$.
- (iv) If $A \subseteq W$, then $A \in \mathcal{F}$ or $W \setminus A \in \mathcal{F}$.

Remark 2.16. Let A be an infinite subset of W . By Zorn's Lemma, there exists a non-principal ultrafilter $\mathcal{F}$ on W such that $A \in \mathcal{F}$.

Definition 2.17. Let $\mathcal{F}$ be a non-principal ultrafilter on W and φ be a property concerning elements of W . We say that φ holds *for almost all* w if

$$\{w \in W \mid \varphi(w) \text{ holds}\} \in \mathcal{F}.$$

We now introduce the notion of ultraproducts, a fundamental construction in model theory and non-standard analysis.

Definition 2.18. Let $(A_w)_{w \in W}$ be a family of non-empty sets indexed by W and $\mathcal{F}$ be a non-principal ultrafilter on W . The *ultraproduct* A_{∞} of $(A_w)_{w \in W}$ is defined as

$$A_{\infty} = \text{ulim}_w A_w := \prod_w A_w / \sim,$$

where $\sim$ is the equivalence relation on $\prod_w A_w$ defined as follows: for any elements $(a_w), (b_w) \in \prod_w A_w$, we have $(a_w) \sim (b_w)$ if and only if $a_w = b_w$ for almost all w . For each $(a_w) \in \prod_w A_w$, the equivalence class of (a_w) in A_{∞} is denoted by $\text{ulim}_w a_w$.

If $A_w = \mathbb{N}$ (resp. $A_w = \mathbb{Q}$) for all $w \in W$, then we write ${}^*\mathbb{N}$ (resp. ${}^*\mathbb{Q}$) to denote $\text{ulim}_w A_w$. An element of ${}^*\mathbb{N}$ (resp. ${}^*\mathbb{Q}$) is called a *non-standard natural number* (resp. *non-standard rational number*).

For a family $(f_w : A_w \rightarrow B_w)_{w \in W}$ of maps, we can construct the ultraproduct $f_{\infty} : A_{\infty} \rightarrow B_{\infty}$ in a natural way. Similarly, we can also define the ultraproduct of binary operators.

Remark 2.19. If $(A_w)_{w \in W}$ is a family of rings, then A_{∞} is naturally equipped with a ring structure. Moreover, if $(M_w)_{w \in W}$ is a family of A_w -modules, then $M_{\infty} := \text{ulim}_w M_w$ carries a natural A_{∞} -module structure.

The following notation will be used throughout this paper.

Notation. Let $\mathcal{F}$ be a non-principal ultrafilter on $\mathcal{P}$.

- (i) The non-standard natural number $\text{ulim}_p p$ is denoted by π .
- (ii) Let $\nu = \text{ulim}_p n_p$ and $\varepsilon = \text{ulim}_p e_p$ be non-standard natural numbers. Then ν^ε is defined to be $\text{ulim}_p n_p^{e_p}$.
- (iii) Let $\tau = \text{ulim}_p t_p$ be a non-standard rational number. Then $\lfloor \tau \rfloor := \text{ulim}_p \lfloor t_p \rfloor$ and $\lceil \tau \rceil := \text{ulim}_p \lceil t_p \rceil$.

The following is a key fact that enables us to apply the theory of ultraproducts to commutative algebra.

Proposition 2.20. *Let $\mathcal{F}$ be a non-principal ultrafilter on $\mathcal{P}$ and $\overline{\mathbb{F}_p}$ be an algebraic closure of the finite field $\mathbb{F}_p$. Then there exists a (non-canonical) field isomorphism*

$$\text{ulim}_p \overline{\mathbb{F}_p} \cong \mathbb{C}.$$

From now on, we fix a non-principal ultrafilter $\mathcal{F}$ on $\mathcal{P}$ and a field isomorphism $\text{ulim}_p \overline{\mathbb{F}_p} \cong \mathbb{C}$. Let $\mathbb{C}[X_1, \dots, X_n]$ (resp. $\overline{\mathbb{F}_p}[X_1, \dots, X_n]$) denote the polynomial ring in n variables over $\mathbb{C}$ (resp. $\overline{\mathbb{F}_p}$).

Theorem 2.21 ([vdD79]). *The natural ring homomorphism*

$$\mathbb{C}[X_1, \dots, X_n] \rightarrow \text{ulim}_p (\overline{\mathbb{F}_p}[X_1, \dots, X_n]); \quad X_i \mapsto \text{ulim}_p X_i,$$

which depends on the choice of the ultrafilter $\mathcal{F}$ and the field isomorphism $\text{ulim}_p \overline{\mathbb{F}_p} \cong \mathbb{C}$, is faithfully flat.

Definition 2.22 ([Sch03, Section 3.2]). (i) Let $f \in \mathbb{C}[X_1, \dots, X_n]$. A family of polynomials $(f_p)_{p \in \mathcal{P}} \in \prod_p (\overline{\mathbb{F}_p}[X_1, \dots, X_n])$ is said to be an *approximation* of f if f is mapped to $\text{ulim}_p f_p$ under the natural homomorphism $\mathbb{C}[X_1, \dots, X_n] \rightarrow \text{ulim}_p (\overline{\mathbb{F}_p}[X_1, \dots, X_n])$.

- (ii) Let $I = (f_1, \dots, f_m)$ be an ideal of $\mathbb{C}[X_1, \dots, X_n]$. A family of ideals $(I_p)_{p \in \mathcal{P}}$ is said to be an *approximation* of I if $I_p = (f_{1,p}, \dots, f_{m,p}) \subseteq \overline{\mathbb{F}_p}[X_1, \dots, X_n]$ for almost all p , where $(f_{i,p})_{p \in \mathcal{P}}$ is an approximation of f_i for each $i = 1, \dots, m$. Note that the definition of approximations of I is independent of the choice of generators $f_1, \dots, f_m$ and of their approximations $(f_{i,p})_{p \in \mathcal{P}}$. More precisely, if $(I_p)_{p \in \mathcal{P}}$ and $(J_p)_{p \in \mathcal{P}}$ are two approximations of I , then $I_p = J_p$ for almost all p .

Proposition 2.23 ([vdD79]). *Let $I = (f_1, \dots, f_m)$ be a prime ideal of $\mathbb{C}[X_1, \dots, X_n]$ and $(I_p)_{p \in \mathcal{P}}$ be an approximation of I . Then $I(\text{ulim}_p (\overline{\mathbb{F}_p}[X_1, \dots, X_n]))$ is a prime ideal of $\text{ulim}_p (\overline{\mathbb{F}_p}[X_1, \dots, X_n])$, and I_p is prime for almost all p .*

As an application of this proposition, we can construct approximations for local rings essentially of finite type over $\mathbb{C}$ as follows.

Definition 2.24 ([Sch03, Section 4.3]). Let R be a local ring essentially of finite type over $\mathbb{C}$, that is, $R \cong (\mathbb{C}[X_1, \dots, X_n]/I)_{\mathfrak{p}}$, where $I \subseteq \mathbb{C}[X_1, \dots, X_n]$ is an ideal and $\mathfrak{p} \subseteq \mathbb{C}[X_1, \dots, X_n]$ is a prime ideal containing I . Let $(I_p)_{p \in \mathcal{P}}$ be an approximation of I and $(\mathfrak{p}_p)_{p \in \mathcal{P}}$ be an approximation of $\mathfrak{p}$. A family $(R_p)_{p \in \mathcal{P}}$ of rings is said to

be an *approximation* of R if $R_p = (\overline{\mathbb{F}_p}[X_1, \dots, X_n]/I_p)_{\mathfrak{p}_p}$ for almost all p , and the ultraproduct $R_\infty := \text{ulim}_p R_p$ is called the *non-standard hull* of R . The non-standard hull is independent of the choice of presentation of R , i.e., any two non-standard hulls are isomorphic as R -algebras.

Remark 2.25. The natural ring homomorphism $R \rightarrow R_\infty$ is known to be flat and local.

Remark 2.26. Approximations of R are closely related to its reduction modulo p . Specifically, the following holds ([Sch04, Corollary 5.9]): Let $x \in X$ be a point of a variety X over $\mathbb{C}$ and $(A, x_A \in X_A)$ be a model of $x \in X$. Suppose that $(R_p)_{p \in \mathcal{P}}$ is an approximation of $R := \mathcal{O}_{X,x}$. For a general closed point $\mu \in A$, the base change $\overline{\{x_\mu\}} \times_{\text{Spec } A/\mu} \text{Spec } \overline{A/\mu}$ is an integral closed subscheme of $X_{\overline{\mu}} := X_\mu \times_{\text{Spec } A/\mu} \text{Spec } \overline{A/\mu}$, where $\overline{A/\mu}$ is an algebraic closure of the finite field A/μ , and we define $x_{\overline{\mu}}$ as its generic point. Then there exists a family of closed points $\{\mu_p\}_{p \in \mathcal{P}}$ of $\text{Spec } A$ such that $R_p \cong \mathcal{O}_{X_{\overline{\mu_p}}, x_{\overline{\mu_p}}}$ for almost all p .

Definition 2.27. (1) Let $f \in R$. A family $(f_p)_{p \in \mathcal{P}}$ of elements is said to be an *approximation* of f if $f_p \in R_p$ for all $p \in \mathcal{P}$ and f is mapped to $\text{ulim}_p f_p$ under the natural map $R \rightarrow R_\infty$.
 (2) Let $I = (f_1, \dots, f_m)$ be an ideal of R , and suppose that $(f_{i,p})_{p \in \mathcal{P}}$ is an approximation of f_i for each $i = 1, \dots, m$. A family $(I_p)_{p \in \mathcal{P}}$ of ideals is said to be an *approximation* of I if $I_p = (f_{1,p}, \dots, f_{m,p})$ for almost all p .

Basic ring-theoretic properties of R are retained by its approximations.

Proposition 2.28 ([Sch03, Theorems 4.5 and 4.6], [Yam23, Proposition 3.9]). *Let $(R_p)_{p \in \mathcal{P}}$ be an approximation of R .*

- (1) *Let d be a natural number. Then R is of dimension d if and only if so is R_p for almost all p .*
- (2) *R is regular (resp. Gorenstein, Cohen–Macaulay, normal) if and only if so is R_p for almost all p .*

For every integer $n \geq 1$, we have a natural homomorphism $R[X_1, \dots, X_n] \rightarrow \text{ulim}_p (R_p[X_1, \dots, X_n])$.

Definition 2.29. Let $(R_p)_{p \in \mathcal{P}}$ be an approximation of R and $R_p[X_1, \dots, X_n]$ (resp. $R[X_1, \dots, X_n]$) denote the polynomial ring in n variables over R_p (resp. R). An *approximation* of a polynomial $f \in R[X_1, \dots, X_n]$ is a family of polynomials $(f_p)_{p \in \mathcal{P}}$ such that $f_p \in R_p[X_1, \dots, X_n]$ for all $p \in \mathcal{P}$ and f is mapped to $\text{ulim}_p f_p$ under the natural ring homomorphism

$$R[X_1, \dots, X_n] \rightarrow \text{ulim}_p (R_p[X_1, \dots, X_n]).$$

An approximation of an ideal in $R[X_1, \dots, X_n]$ is defined in an analogous way.

Definition 2.30. Let S be a ring of finite type over R , that is, $S \cong R[X_1, \dots, X_n]/I$ for some ideal $I \subseteq R[X_1, \dots, X_n]$.

- (i) A family of rings $(S_p)_{p \in \mathcal{P}}$ is said to be a *relative approximation* of S if $S_p = R_p[X_1, \dots, X_n]/I_p$ for almost all p , where $(I_p)_{p \in \mathcal{P}}$ is an approximation of I . The ultraproduct $S_\infty := \text{ulim}_p S_p$ is called the *relative hull* of S .
- (ii) Let M be a finitely generated S -module. We can write M as the cokernel of an S -linear map between finite free modules, i.e., there exists an exact sequence

$$S^m \xrightarrow{A} S^n \rightarrow M \rightarrow 0$$

such that A is an $n \times m$ -matrix with entries in S . Let $(S_p)_{p \in \mathcal{P}}$ be an approximation of S and $(A_p)_{p \in \mathcal{P}}$ be an entrywise approximation of A . A family $(M_p)_{p \in \mathcal{P}}$ of S_p -modules is said to be an *approximation* of M if M_p is the cokernel of the S_p -linear map $S_p^m \xrightarrow{A_p} S_p^n$ for almost all p .

Remark 2.31. The S_∞ -module $\text{ulim}_p M_p$ is isomorphic to $M \otimes_S S_\infty$. This isomorphism does not depend on the choice of presentation of M .

We next introduce ultra-Frobenii, which are an ultraproduct analog of Frobenius maps.

Definition 2.32. Let S be a ring of finite type over R .

- (i) Let $\varepsilon = \text{ulim}_p e_p$ be a non-standard natural number. An *ultra-Frobenius* $F^\varepsilon : S \rightarrow S_\infty$ is the ring homomorphism that sends x to $\text{ulim}_p x_p^{e_p}$, where $(x_p)_{p \in \mathcal{P}}$ is an approximation of x . We also use the same notion F^ε to denote the ring homomorphism $S_\infty \rightarrow S_\infty$ that sends $\text{ulim}_p x_p$ to $\text{ulim}_p x_p^{e_p}$.
- (ii) For an element $f = \text{ulim}_p f_p \in S_\infty$ and a non-standard natural number ε , we define $f^\varepsilon := \text{ulim}_p f_p^{e_p}$. Note that if ε is a standard natural number, then f^ε agrees with the usual power of f .
- (iii) For a non-standard natural number ε , let F_*^ε denote the restriction of scalars along $F^\varepsilon : S_\infty \rightarrow S_\infty$. For an S_∞ -module M and an element $x \in M$, we write $F_*^\varepsilon x$ to denote x viewed as an element of $F_*^\varepsilon M$.

Suppose that $(R, \mathfrak{m})$ is a d -dimensional local ring essentially of finite type over $\mathbb{C}$ and $x_1, \dots, x_d$ is a system of parameters for R . Let $(R_p)_{p \in \mathcal{P}}$ and $(x_p)_{p \in \mathcal{P}}$ be approximations of R and $x := x_1 \cdots x_d$, respectively, and let $(M_p)_{p \in \mathcal{P}}$ be a family of R_p -modules. The Čech complex of M_p induces a short exact sequence

$$\bigoplus_{i=1}^d (M_p)_{x_{i,p}} \rightarrow (M_p)_{x_p} \rightarrow H_{\mathfrak{m}_p}^d(M_p) \rightarrow 0,$$

where $(x_{i,p})_{p \in \mathcal{P}}$ is an approximation of $\check{x}_i := x_1 \cdots x_{i-1} x_{i+1} \cdots x_d$, for almost all p . Taking ultraproducts yields an exact sequence

$$\bigoplus_{i=1}^d \text{ulim}_p (M_p)_{x_{i,p}} \rightarrow \text{ulim}_p (M_p)_{x_p} \rightarrow \text{ulim}_p H_{\mathfrak{m}_p}^d(M_p) \rightarrow 0$$

Then, from the commutative diagram

$$\begin{array}{ccc} \bigoplus_{i=1}^d (\operatorname{ulim}_p M_p)_{\tilde{x}_i} & \longrightarrow & (\operatorname{ulim}_p M_p)_x, \\ \downarrow & & \downarrow \\ \bigoplus_{i=1}^d \operatorname{ulim}_p (M_p)_{x_{i,p}} & \longrightarrow & \operatorname{ulim}_p (M_p)_{x_p} \end{array}$$

we have a canonical homomorphism

$$H_{\mathfrak{m}}^d(\operatorname{ulim}_p M_p) \rightarrow \operatorname{ulim}_p H_{\mathfrak{m}_p}^d(M_p).$$

For an element

$$\eta = \left[\frac{y}{x^t} \right] \in H_{\mathfrak{m}}^d(\operatorname{ulim}_p M_p) \quad (y \in \operatorname{ulim}_p M_p, t \in \mathbb{N}),$$

a family $(\eta_p)_{p \in \mathcal{P}}$ of elements of $H_{\mathfrak{m}_p}^d(M_p)$ is called an *approximation* of η if

$$\eta_p = \left[\frac{y_p}{x_p^t} \right] \in H_{\mathfrak{m}_p}^d(M_p)$$

for almost all p . If $(\eta_p)_{p \in \mathcal{P}}$ is an approximation of η , then the image of η under the above canonical homomorphism is equal to $\operatorname{ulim}_p \eta_p$.

3. LOCAL F -ALPHA INVARIANTS

In this section, we introduce a local version of Pande's F -alpha invariants (see [Pan23])—a positive characteristic analog of Tian's alpha invariants—and study its basic properties.

3.1. Definition and basic properties.

Definition 3.1. Let $(R, \mathfrak{m})$ be a Noetherian local domain and $x \in R$ be a nonzero element of R . The *normalized order* $\overline{\operatorname{ord}}_{\mathfrak{m}}(x)$ of x with respect to the maximal ideal $\mathfrak{m}$ is defined as

$$\overline{\operatorname{ord}}_{\mathfrak{m}}(x) = \max\{n \geq 0 \mid x \in \overline{\mathfrak{m}^n}\},$$

where $\overline{\mathfrak{m}^n}$ is the integral closure of the ideal $\mathfrak{m}^n$.

Remark 3.2. (1) $\overline{\operatorname{ord}}_{\mathfrak{m}}(x^\ell) = \ell \overline{\operatorname{ord}}_{\mathfrak{m}}(x)$ for all integers $\ell \geq 1$.

(2) If R is a regular local ring or the localization of a normal standard graded ring over a field at the unique homogeneous maximal ideal, then [Pol26, Proposition 2.10] shows that

$$\overline{\operatorname{ord}}_{\mathfrak{m}}(x) = \operatorname{ord}_{\mathfrak{m}}(x) := \max\{n \geq 0 \mid x \in \mathfrak{m}^n\}.$$

We now introduce the notion of local F -alpha invariants, which serve as a local analog of Pande's F -alpha invariants [Pan23].

Definition 3.3. Let $(R, \mathfrak{m})$ be a Noetherian F -finite F -pure local domain of characteristic $p > 0$ and of dimension $d \geq 1$. The *local F -alpha invariant* $\alpha_F(R)$ of R is defined as

$$\alpha_F(R) = \inf_{f \in \mathfrak{m}} \operatorname{fpt}(f) \overline{\operatorname{ord}}_{\mathfrak{m}}(f),$$

where f runs through all nonzero elements of $\mathfrak{m}$.

Remark 3.4. If R is not strongly F -regular, then $\alpha_F(R) = 0$. Indeed, by the definition of strong F -regularity, there exists a nonzero element $c \in \mathfrak{m}$ such that for every $e \geq 1$, the map $R \xrightarrow{\cdot F_*^e c} F_*^e R$ fails to split. Therefore, $\text{fpt}(c) = 0$, which shows that $\alpha_F(R) = 0$.

Proposition 3.5. *Let $(R, \mathfrak{m})$ be a strongly F -regular local domain of characteristic $p > 0$ and of dimension $d \geq 1$.*

- (1) $0 \leq \alpha_F(R) \leq 1$.
- (2) If R is regular, then $\alpha_F(R) = 1$.
- (3) Let $\widehat{R}$ denote the $\mathfrak{m}$ -adic completion of R . Then $\alpha_F(R) = \alpha_F(\widehat{R})$.

Proof. (1) Fix an element $x \in \mathfrak{m} \setminus \overline{\mathfrak{m}}^2$. Then $\alpha_F(R) \leq \text{fpt}(x) \leq 1$.

(2) If R is regular, then for every nonzero element $f \in \mathfrak{m}$, we have

$$\text{fpt}(f) \overline{\text{ord}}_{\mathfrak{m}}(f) = \text{fpt}(f) \text{ord}_{\mathfrak{m}}(f) \geq 1$$

by [TW04, Proposition 4.1]. Thus, $\alpha_F(R) = 1$.

(3) For every nonzero element $f \in \mathfrak{m}$, we have the equalities $\text{fpt}(R, f) = \text{fpt}(\widehat{R}, f)$ (see Remark 2.4) and $\overline{\text{ord}}_{\mathfrak{m}}(f) = \overline{\text{ord}}_{\mathfrak{m}\widehat{R}}(f)$ (see [HS06, Proposition 1.6.2]). Therefore, $\alpha_F(R) \geq \alpha_F(\widehat{R})$.

Suppose that $\alpha_F(R) > \alpha_F(\widehat{R})$. Let $\varepsilon > 0$ be a real number such that $\varepsilon < \alpha_F(R) - \alpha_F(\widehat{R})$. By the definition of $\alpha_F(\widehat{R})$, there exists a nonzero element $f \in \mathfrak{m}\widehat{R}$ such that $\text{fpt}(\widehat{R}, f) \overline{\text{ord}}_{\mathfrak{m}\widehat{R}}(f) < \alpha_F(R) - \varepsilon$. Fix a sufficiently large integer N such that

$$\text{fpt}(\widehat{R}, \mathfrak{m}\widehat{R}) \overline{\text{ord}}_{\mathfrak{m}\widehat{R}}(f) < N\varepsilon \quad \text{and} \quad \overline{\text{ord}}_{\mathfrak{m}\widehat{R}}(f) < N.$$

Take an element $g \in R$ such that $f - g \in \mathfrak{m}^N \widehat{R}$. Then by [TW04, Proposition 4.4],

$$\begin{aligned} \text{fpt}(R, g) &= \text{fpt}(\widehat{R}, g) \leq \text{fpt}(\widehat{R}, f) + \text{fpt}(\widehat{R}, \mathfrak{m}^N \widehat{R}) \\ &< \text{fpt}(\widehat{R}, f) + \frac{\varepsilon}{\overline{\text{ord}}_{\mathfrak{m}\widehat{R}}(f)}. \end{aligned}$$

Since $\overline{\text{ord}}_{\mathfrak{m}}(g) = \overline{\text{ord}}_{\mathfrak{m}\widehat{R}}(f)$ by the choice of g , it follows that

$$\text{fpt}(R, g) \overline{\text{ord}}_{\mathfrak{m}}(g) < \text{fpt}(\widehat{R}, f) \overline{\text{ord}}_{\mathfrak{m}\widehat{R}}(f) + \varepsilon < \alpha_F(R),$$

which contradicts the definition of $\alpha_F(R)$. $\square$

Next, we present a lemma on F -pure thresholds that will be used to give an alternative description of local F -alpha invariants.

Lemma 3.6 (cf. [MTW05]). *Let $(R, \mathfrak{m})$ be a strongly F -regular local domain of characteristic $p > 0$, and let $\mathfrak{a}$ be a nonzero ideal of R contained in $\mathfrak{m}$.*

- (1) For each integer $e \geq 1$, set

$$\nu_e(\mathfrak{a}) = \max\{r \geq 0 \mid F_*^e \mathfrak{a}^r \cdot \text{Hom}_R(F_*^e R, R) \rightarrow R \text{ is surjective}\},$$

where the map

$$F_*^e \mathfrak{a}^r \cdot \text{Hom}_R(F_*^e R, R) \rightarrow R$$

is the evaluation map sending $F_*^e x \cdot \varphi$ to $\varphi(F_*^e x)$. Here we view $\mathrm{Hom}_R(F_*^e R, R)$ as an $F_*^e R$ -module, so that $F_*^e \mathfrak{a}^r \cdot \mathrm{Hom}_R(F_*^e R, R)$ is an $F_*^e R$ -submodule of $\mathrm{Hom}_R(F_*^e R, R)$. Then the sequence $(\nu_e(\mathfrak{a})/p^e)_{e \geq 1}$ is monotonically increasing and

$$\mathrm{fpt}(\mathfrak{a}) = \sup_e \frac{\nu_e(\mathfrak{a})}{p^e}.$$

Moreover, for all integers $e \geq 1$,

$$\mathrm{fpt}(\mathfrak{a}) > \frac{\nu_e(\mathfrak{a})}{p^e}.$$

(2) Suppose that $\mathfrak{a} = (f)$ is a principal ideal. For each integer $e \geq 1$, set

$$\begin{aligned} \mu_e(f) &= \nu_e(f) + 1 \\ &= \min\{r \geq 0 \mid F_*^e f^r \cdot \mathrm{Hom}_R(F_*^e R, R) \rightarrow R \text{ is not surjective}\}. \end{aligned}$$

Then the sequence $(\mu_e(f)/p^e)_{e \geq 1}$ is monotonically decreasing and

$$\mathrm{fpt}(f) = \inf_e \frac{\mu_e(f)}{p^e}.$$

Proof. The proof closely follows that of [MTW05], although the ring is assumed to be regular in *loc. cit.* First, recall from [TT08, Proposition 4.4] that $\mathrm{fpt}(\mathfrak{a}) = \lim_{e \rightarrow \infty} \nu_e(\mathfrak{a})/p^e$ and $\mathrm{fpt}(f) = \lim_{e \rightarrow \infty} \mu_e(f)/p^e$.

For the first part of (1), it suffices to show that $p\nu_e(\mathfrak{a}) \leq \nu_{e+1}(\mathfrak{a})$ for every integer $e \geq 1$. For (2), it suffices to show that $p\mu_e(\mathfrak{a}) \geq \mu_{e+1}(\mathfrak{a})$ for every integer $e \geq 1$. These inequalities are an immediate consequence of the following elementary fact: for any ideal $\mathfrak{b}$ in R , the map $F_*^{e+1} \mathfrak{b}^{[p]} \cdot \mathrm{Hom}_R(F_*^{e+1} R, R) \rightarrow R$ factors through $F_*^e \mathfrak{b} \cdot \mathrm{Hom}_R(F_*^e R, R) \rightarrow R$, and since R is F -pure, the map

$$F_*^{e+1} \mathfrak{b}^{[p]} \cdot \mathrm{Hom}_R(F_*^{e+1} R, R) \rightarrow F_*^e \mathfrak{b} \cdot \mathrm{Hom}_R(F_*^e R, R)$$

is surjective.

For the second part of (1), it suffices to show that there exists an integer $e_0 \geq 1$ such that $p^{e_0} \nu_e(\mathfrak{a}) < \nu_{e+e_0}(\mathfrak{a})$ for every integer $e \geq 1$. Since R is strongly F -regular, there exists an integer $e_0 \geq 1$ such that $F_*^{e_0} \mathfrak{a} \cdot \mathrm{Hom}_R(F_*^{e_0} R, R) \rightarrow R$ is surjective. Then for all integers $e \geq 1$ and $r \geq 0$, the map

$$F_*^{e+e_0} \mathfrak{a}^{p^{e_0} r + 1} \cdot \mathrm{Hom}_R(F_*^{e+e_0} R, R) \rightarrow F_*^e \mathfrak{a}^r \cdot \mathrm{Hom}_R(F_*^e R, R)$$

is surjective. This implies the desired inequality. $\square$

Proposition 3.7. *Let $(R, \mathfrak{m})$ be a strongly F -regular local domain of characteristic $p > 0$ and of dimension $d \geq 1$. For every integer $e \geq 1$, consider the ideal $I_e(R) \subseteq R$ defined by*

$$I_e = I_e(R) := \{c \in R \mid F_*^e c \cdot \mathrm{Hom}_R(F_*^e R, R) \rightarrow R \text{ is not surjective}\}.$$

Let $m_e(R)$ be the largest integer m such that $I_e(R) \subseteq \overline{\mathfrak{m}}^m$. Then

$$\alpha_F(R) = \lim_{e \rightarrow \infty} \frac{m_e(R)}{p^e}.$$

Proof. First, we show that $\alpha_F(R) \leq \liminf_{e \rightarrow \infty} m_e(R)/p^e$. For every integer $e \geq 1$, by the definition of $m_e(R)$, there exists an element $c_e \in I_e$ not contained in $\mathfrak{m}^{\overline{m_e(R)+1}}$. Then $\mu_e(c_e) = 1$ and $\overline{\text{ord}}_{\mathfrak{m}}(c_e) = m_e(R)$. It follows from Lemma 3.6 (2) that

$$\text{fpt}(c_e) \overline{\text{ord}}_{\mathfrak{m}}(c_e) \leq \frac{m_e(R)}{p^e},$$

which implies the desired inequality.

Next, we show that $\alpha_F(R) \geq \limsup_{e \rightarrow \infty} m_e(R)/p^e$. Fix a nonzero element $f \in \mathfrak{m}$. Since $f^{\mu_e(f)} \in I_e \subseteq \overline{\mathfrak{m}^{m_e(R)}}$ by definition, we have

$$\mu_e(f) \overline{\text{ord}}_{\mathfrak{m}}(f) = \overline{\text{ord}}_{\mathfrak{m}}(f^{\mu_e(f)}) \geq m_e(R).$$

Dividing both sides by p^e and taking the limit superior, we obtain

$$\text{fpt}(f) \overline{\text{ord}}_{\mathfrak{m}}(f) \geq \limsup_{e \rightarrow \infty} \frac{m_e(R)}{p^e}.$$

□

Remark 3.8. F -signature can be described in terms of the ideals I_e . It follows from [Tuc12, Proposition 4.5] that $s(R) = \lim_{e \rightarrow \infty} \ell_R(R/I_e)/p^{de}$. We use this description in the following proposition.

We follow Pande's argument to establish basic bounds for local F -alpha invariants.

Proposition 3.9 (cf. [Pan23, Theorem 4.6]). *Let $(R, \mathfrak{m})$ be a strongly F -regular local domain of characteristic $p > 0$ and of dimension $d \geq 1$. Let $e(R)$ denote the Hilbert–Samuel multiplicity of R .*

(1) *The following inequalities hold:*

$$\frac{e(R) \alpha_F(R)^d}{d!} \leq s(R) \leq \frac{e(R) \text{fpt}(\mathfrak{m})^d}{d!}.$$

(2) *Let $C := \text{edim}(R) + 1$ and let w be a Rees valuation centered at the maximal ideal $\mathfrak{m}$. By Izumi's theorem (see [HS01, Theorem 1.2]), there exists an integer $N \geq 1$ such that*

$$w(f) \leq N \overline{\text{ord}}_{\mathfrak{m}}(f) \leq N w(f)$$

for every element $f \in R$. Then

$$s(R) \leq \frac{e(R) N^d}{d!} (C^d - (C - \alpha_F(R))^d).$$

Proof. (1) First, we show the upper bound for $s(R)$. By Lemma 3.6 (1), we have $\text{fpt}(\mathfrak{m}) p^e > \nu_e(\mathfrak{m})$, so that $\mathfrak{m}^{\lceil \text{fpt}(\mathfrak{m}) p^e \rceil} \subseteq I_e$. Then by Remark 3.8,

$$\begin{aligned} s(R) &= \lim_{e \rightarrow \infty} \frac{\ell_R(R/I_e)}{p^{de}} \leq \lim_{e \rightarrow \infty} \frac{\ell_R(R/\mathfrak{m}^{\lceil \text{fpt}(\mathfrak{m}) p^e \rceil})}{p^{de}} \\ &= \lim_{e \rightarrow \infty} \frac{1}{d!} \frac{\ell_R(R/\mathfrak{m}^{\lceil \text{fpt}(\mathfrak{m}) p^e \rceil}) d!}{\lceil \text{fpt}(\mathfrak{m}) p^e \rceil^d} \left(\frac{\lceil \text{fpt}(\mathfrak{m}) p^e \rceil}{p^e} \right)^d \\ &= \frac{1}{d!} e(R) \text{fpt}(\mathfrak{m})^d. \end{aligned}$$

For the lower bound for $s(R)$, if $\alpha_F(R) = 0$, then the inequality is clear. Hence, we may assume that $\alpha_F(R) > 0$, and then, by Proposition 3.7, we have $m_e(R) \rightarrow \infty$ as $e \rightarrow \infty$. Note that the zeroth normal Hilbert coefficient

$$\bar{e}_0(\mathfrak{m}) = \lim_{n \rightarrow \infty} d! \frac{\ell_R(R/\overline{\mathfrak{m}^n})}{n^d}$$

of $\mathfrak{m}$ is nothing but the Hilbert–Samuel multiplicity $e(R)$ of R . Then, using Remark 3.8 again, we have

$$\begin{aligned} s(R) &= \lim_{e \rightarrow \infty} \frac{\ell_R(R/I_e)}{p^{de}} \geq \lim_{e \rightarrow \infty} \frac{\ell_R(R/\overline{\mathfrak{m}^{m_e(R)}})}{p^{de}} \\ &= \lim_{e \rightarrow \infty} \frac{1}{d!} \frac{\ell_R(R/\overline{\mathfrak{m}^{m_e(R)}}) d!}{m_e(R)^d} \left(\frac{m_e(R)}{p^e} \right)^d \\ &= \frac{1}{d!} e(R) \alpha_F(R)^d. \end{aligned}$$

(2) Let $\pi : Y \rightarrow \operatorname{Spec} R$ be the normalized blow-up of $\mathfrak{m}$ and write

$$\mathfrak{m}\mathcal{O}_Y = \mathcal{O}_Y(-\sum_{i=1}^r a_i E_i),$$

where the E_i are distinct prime Weil divisors on Y and the a_i are positive integers. We may assume that $w = \operatorname{ord}_{E_1}/a_1$. By Izumi's theorem [HS01, Theorem 1.2] and the description of integral closure in terms of Rees valuations (see the paragraph preceding [HS01, Theorem 2.6]; cf. [BFJ14, Theorem 4.3 (i)]), there exists a positive integer N such that for every $f \in R$, we have

$$w(f) \leq N \overline{\operatorname{ord}}_{\mathfrak{m}}(f) \leq N w(f).$$

For each rational number $t \geq 0$, let $\mathfrak{a}_t := \{f \in R \mid w(f) \geq t\}$, and define the invariant $\alpha_{F,w}(R)$ associated with w by

$$\alpha_{F,w}(R) := \inf_{f \in \mathfrak{m}} \operatorname{fpt}(f) w(f),$$

where f runs through all nonzero elements of $\mathfrak{m}$. It is easy to see that $\mathfrak{a}_t$ is an ideal of R and that $0 \leq \alpha_{F,w}(R) \leq \min\{1, N\alpha_F(R)\}$.

Let $0 < \varepsilon < 1$ be arbitrary. By the definition of $\alpha_{F,w}(R)$, there exists a nonzero element $f \in \mathfrak{m}$ such that $\operatorname{fpt}(f) w(f) < \alpha_{F,w}(R) + \varepsilon$. Then we can choose positive integers a and e_0 such that

$$\operatorname{fpt}(f) < \frac{a}{p^{e_0} - 1} < \frac{\alpha_{F,w}(R) + \varepsilon}{w(f)}.$$

Replacing f by f^a , we may assume that $a = 1$. For each integer $s \geq w(f)$, set

$$v_s := \frac{p^{se_0} - 1}{p^{e_0} - 1}.$$

Since $1/(p^{e_0} - 1) > \text{fpt}(f)$, it follows from [Pan23, Lemma 3.11] that $f^{v_s} \in I_{se_0}(R)$. Moreover, for all sufficiently large e , the Briançon–Skoda theorem gives

$$\overline{\mathfrak{m}^{Cp^e}} \subseteq \mathfrak{m}^{\text{edim}(R)p^e} \subseteq \mathfrak{m}^{[p^e]} \subseteq I_e(R).$$

For $s \gg 0$, we have

$$NCp^{se_0} \geq 2p^{se_0} > (\alpha_{F,w}(R) + \varepsilon)p^{se_0} > \frac{w(f)}{p^{e_0} - 1}p^{se_0} > w(f)v_s.$$

Setting $M(s) := NCp^{se_0} - w(f)v_s > 0$, we consider the exact sequence

$$0 \rightarrow R/\mathfrak{a}_{M(s)} \xrightarrow{\cdot f^{v_s}} R/\mathfrak{a}_{NCp^{se_0}} \rightarrow R/(f^{v_s}R + \mathfrak{a}_{NCp^{se_0}}) \rightarrow 0.$$

By the choice of N and the containments above, we have

$$f^{v_s}R + \mathfrak{a}_{NCp^{se_0}} \subseteq f^{v_s}R + \overline{\mathfrak{m}^{Cp^{se_0}}} \subseteq I_{se_0}(R).$$

Together with the exact sequence above, this gives

$$\begin{aligned} \ell_R(R/I_{se_0}(R)) &\leq \ell_R(R/(f^{v_s}R + \mathfrak{a}_{NCp^{se_0}})) \\ &= \ell_R(R/\mathfrak{a}_{NCp^{se_0}}) - \ell_R(R/\mathfrak{a}_{M(s)}). \end{aligned}$$

Therefore,

$$\begin{aligned} s(R) &= \lim_{s \rightarrow \infty} \frac{1}{p^{se_0d}} (\ell_R(R/I_{se_0}(R))) \\ &\leq \lim_{s \rightarrow \infty} \frac{1}{p^{se_0d}} (\ell_R(R/\mathfrak{a}_{NCp^{se_0}}) - \ell_R(R/\mathfrak{a}_{M(s)})) \\ &= \frac{\text{vol}(w)}{d!} \left(N^d C^d - \left(NC - \frac{w(f)}{p^{e_0} - 1} \right)^d \right) \\ &\leq \frac{\text{vol}(w)}{d!} (N^d C^d - (NC - (\alpha_{F,w}(R) + \varepsilon))^d) \\ &\leq \frac{e(R)}{d!} (N^d C^d - (NC - (\alpha_{F,w}(R) + \varepsilon))^d). \end{aligned}$$

Here $\text{vol}(w)$ denotes the volume of the valuation w , defined by

$$\text{vol}(w) := \lim_{t \rightarrow \infty} \frac{d!}{t^d} \ell_R(R/\mathfrak{a}_t),$$

as in [ELS03]; see also [Cut13]. Letting ε tend to zero yields

$$\begin{aligned} s(R) &\leq \frac{e(R)}{d!} (N^d C^d - (NC - \alpha_{F,w}(R))^d) \\ &\leq \frac{e(R)N^d}{d!} (C^d - (C - \alpha_F(R))^d), \end{aligned}$$

where the last inequality follows from the inequality $\alpha_{F,w}(R) \leq N\alpha_F(R)$. $\square$

Remark 3.10. Suppose that $(R, \mathfrak{m})$ is a local ring essentially of finite type over $\mathbb{C}$ whose spectrum is of klt type. By the proof of Lemma 4.14, the integer N in Proposition 3.9 (2) for a reduction R_μ of R modulo p can be chosen uniformly for general closed points μ .

Remark 3.11. Let $(R, \mathfrak{m})$ be a Noetherian F -finite F -pure local domain of characteristic $p > 0$ and of dimension $d \geq 1$. Remark 3.4 and Proposition 3.9 imply that R is strongly F -regular if and only if $\alpha_F(R) > 0$.

3.2. Local versus graded settings. In this subsection, we compare our local invariants with the graded invariants introduced by Pande. Throughout this subsection, we assume that $R = \bigoplus_{i \geq 0} R_i$ is a strongly F -regular graded domain of dimension $d \geq 1$ and of finite type over an F -finite field $R_0 = k$ of characteristic $p > 0$, and let $\mathfrak{m} = \bigoplus_{i \geq 1} R_i$ denote the irrelevant ideal of R .

Pande [Pan23] defined “ α_F -invariants” for section rings, but her definition applies more generally to graded rings over an F -finite field.

Definition 3.12 (cf. [Pan23, Definition 3.4]). Let Δ be an effective homogeneous $\mathbb{Q}$ -Weil divisor on $\text{Spec } R$. Then the F -alpha invariant $\alpha_F(R, \Delta)$ of the pair (R, Δ) is defined as

$$\alpha_F(R, \Delta) = \inf_f \text{fpt}((R, \Delta); f) \deg(f),$$

where f runs through all nonzero homogeneous elements of $\mathfrak{m}$. If $\Delta = 0$, then this invariant is said to be the F -alpha invariant of R and simply denoted by $\alpha_F(R)$.

To study the properties of F -alpha invariants, we consider the graded version of the ideals $I_e(R)$ introduced in Proposition 3.7.

Definition 3.13. For every integer $e \geq 1$, the ideal $I_e(R) \subseteq R$ is defined as

$$I_e(R) = \{x \in R \mid \varphi(x) \in \mathfrak{m} \text{ for all } \varphi \in \text{Hom}_R(F_*^e R, R)\}.$$

Remark 3.14. (1) (cf. [DSNB18, Proposition 3.7]) $I_e(R)$ is a homogeneous ideal.

(2) (cf. [DSNB18, Proposition 3.10]) F -pure thresholds of R can be described in terms of $I_e(R)$ just as in the local case (see Lemma 3.6). Let $\mathfrak{a}$ be a nonzero homogeneous ideal of R . For each integer $e \geq 1$, define

$$\nu_e(f) = \max\{r \geq 0 \mid \mathfrak{a}^r \not\subseteq I_e(R)\}.$$

Then

$$\text{fpt}(\mathfrak{a}) = \lim_{e \rightarrow \infty} \frac{\nu_e(\mathfrak{a})}{p^e}.$$

(3) $I_e(R) = I_e(R_{\mathfrak{m}}) \cap R$, where the definition of $I_e(R_{\mathfrak{m}})$ can be found in Proposition 3.7.

Lemma 3.15. Let $f \in \mathfrak{m}$ be a nonzero element and f_0 denote the lowest nonzero homogeneous component of f . Then $\text{fpt}(R_{\mathfrak{m}}, f) \geq \text{fpt}(R_{\mathfrak{m}}, f_0)$.

Proof. For every integer $e \geq 1$ and every nonzero element $g \in \mathfrak{m}$, set

$$\nu_e(R_{\mathfrak{m}}, g) := \max\{r \geq 0 \mid g^r \notin I_e(R_{\mathfrak{m}})\}.$$

Note that $f_0^{\nu_e(R_{\mathfrak{m}}, f_0)} \notin I_e(R_{\mathfrak{m}}) \cap R = I_e(R)$. Since $I_e(R)$ is a homogeneous ideal and $f_0^{\nu_e(R_{\mathfrak{m}}, f_0)}$ is a homogeneous component of $f^{\nu_e(R_{\mathfrak{m}}, f_0)}$, we see that $f^{\nu_e(R_{\mathfrak{m}}, f_0)} \notin I_e(R)$, that is, $\nu_e(R_{\mathfrak{m}}, f) \geq \nu_e(R_{\mathfrak{m}}, f_0)$. It then follows from Proposition 3.6 that

$$\text{fpt}(R_{\mathfrak{m}}, f_0) = \lim_{e \rightarrow \infty} \frac{\nu_e(R_{\mathfrak{m}}, f_0)}{p^e} \leq \lim_{e \rightarrow \infty} \frac{\nu_e(R_{\mathfrak{m}}, f)}{p^e} = \text{fpt}(R_{\mathfrak{m}}, f).$$

□

Proposition 3.16. *Suppose that R is standard graded. Then*

$$\alpha_F(R) = \alpha_F(R_{\mathfrak{m}}).$$

Proof. It follows that

$$\begin{aligned} \alpha_F(R_{\mathfrak{m}}) &= \inf_{f \in \mathfrak{m}} \text{fpt}(R_{\mathfrak{m}}, f) \overline{\text{ord}}_{\mathfrak{m}R_{\mathfrak{m}}}(f) \\ &= \inf_{g \in \mathfrak{m} : \text{homog}} \text{fpt}(R_{\mathfrak{m}}, g) \overline{\text{ord}}_{\mathfrak{m}R_{\mathfrak{m}}}(g) \\ &= \inf_{g \in \mathfrak{m} : \text{homog}} \text{fpt}(R_{\mathfrak{m}}, g) \deg(g) \\ &= \alpha_F(R), \end{aligned}$$

where g runs through all nonzero homogeneous elements of $\mathfrak{m}$. Here, the second and third equalities use Lemma 3.15 and the formula $\overline{\text{ord}}_{\mathfrak{m}R_{\mathfrak{m}}}(g) = \text{ord}_{\mathfrak{m}R_{\mathfrak{m}}}(g) = \deg(g)$ given in [Pol26, Proposition 2.10]. The fourth equality follows from the fact that $\text{fpt}(R_{\mathfrak{m}}, g) = \text{fpt}(R, g)$ for every nonzero homogeneous element $g \in \mathfrak{m}$, which is a consequence of Lemma 3.6 and Remark 3.14. □

Proposition 3.17 (cf. [Pan23, Proposition 4.12]). *For every integer $n \geq 1$, let $R^{(n)} = \bigoplus_{i \geq 0} R_{in}$ denote the n th Veronese subring of R . Then*

$$\alpha_F(R) \leq n\alpha_F(R^{(n)}).$$

Moreover, if n is sufficiently divisible, then $n\alpha_F(R^{(n)})$ stabilizes.

Proof. Let $\mathfrak{m}^{(n)} = \bigoplus_{i \geq 1} R_{in}$ denote the irrelevant ideal of $R^{(n)}$. To prove the first assertion, it suffices to show that $\text{fpt}(R, f) \leq \text{fpt}(R^{(n)}, f)$ for every nonzero homogeneous element $f \in \mathfrak{m}^{(n)}$. By Remark 3.14, it is enough to verify that for any positive integers a, e , if $f^a \in I_e(R^{(n)})$, then $f^a \in I_e(R)$. Let $\iota : R^{(n)} \rightarrow R$ be the natural inclusion and $\pi : R \rightarrow R^{(n)}$ the projection. Consider the map

$$\Phi : \text{Hom}_R(F_*^e R, R) \rightarrow \text{Hom}_{R^{(n)}}(F_*^e R^{(n)}, R^{(n)}); \quad \varphi \longmapsto \pi \circ \varphi \circ F_*^e \iota.$$

If $f^a \in I_e(R^{(n)})$, then $\Phi(\varphi)(F_*^e f^a) \in \mathfrak{m}^{(n)}$ for all $\varphi \in \text{Hom}_R(F_*^e R, R)$. This implies that $\varphi(F_*^e f^a) \in \mathfrak{m}$, therefore $f^a \in I_e(R)$.

Next, we prove the latter assertion. By the Pinkham–Demazure construction (see, for example, [Wat81]), there exists an ample $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -Weil divisor D on $X := \text{Proj } R$ such that R is isomorphic to the graded ring $\bigoplus_{i \geq 0} H^0(X, \mathcal{O}_X([iD]))$. Choose

an integer $N \geq 1$ such that ND is an ample Cartier divisor. Then, for every integer $l \geq 1$, it follows from [Pan23, Proposition 4.10] that $l\alpha_F(R^{(lN)}) = \alpha_F(R^{(N)})$, which completes the proof. $\square$

In Proposition 3.17, the equality does not hold in general.

Example 3.18. Let k be a perfect field of characteristic $p > 5$ and

$$R = k[X, Y, Z]/(X^2 + Y^3 + Z^5)$$

be a graded k -algebra with $\deg x = 15$, $\deg y = 10$, $\deg z = 6$, where x, y, z denote the images of X, Y, Z , respectively. Since $\text{fpt}(R, z) = 1/6$, we have

$$\alpha_F(R) \leq \text{fpt}(R, z) \deg z = \frac{1}{6} \cdot 6 = 1.$$

On the other hand, the second Veronese subring $R^{(2)}$ is the polynomial ring $k[y, z]$, so $\text{fpt}(R^{(2)}, z) = 1$. Therefore, $\alpha_F(R^{(2)}) \leq \text{fpt}(R^{(2)}, z) \deg(z) = 1 \cdot 3 = 3$.

We now show the converse inequality. Let f be a nonzero homogeneous element of $\mathfrak{m}^{(2)}$ and $t \in (0, 3/\deg f)$ be any real number. For all sufficiently large $e \gg 0$, we see that

$$\deg f \lfloor t(p^e - 1) \rfloor \leq 3(p^e - 1),$$

which implies that $f^{\lfloor t(p^e - 1) \rfloor} \notin (y, z)^{[p^e]} R^{(2)} \subseteq I_e(R^{(2)})$. It then follows from the Fedder type criterion (see, for example, [Tak04a, Lemma 3.9]) that $\text{fpt}(R^{(2)}, f) \geq t$. Since t is arbitrary in the interval $(0, 3/\deg f)$, we conclude that $\text{fpt}(R^{(2)}, f) \deg f \geq 3$, and hence $\alpha_F(R^{(2)}) = 3$. Thus, $\alpha_F(R) \leq 1 < 6 = 2\alpha_F(R^{(2)})$.

We also introduce F -alpha invariants for projective varieties.

Definition 3.19 (cf. [Pan23, Definition 5.2]). Suppose that (X, Δ) is a globally F -regular pair, where X is a normal projective variety over an F -finite field of characteristic $p > 0$ and Δ is an effective $\mathbb{Q}$ -Weil divisor on X . Given a big $\mathbb{Q}$ -Weil divisor Γ on X , we define the F -alpha invariant $\alpha_F((X, \Delta); \Gamma)$ of (X, Δ) with respect to Γ as

$$\alpha_F((X, \Delta); \Gamma) = \inf_D \text{gst}((X, \Delta); D),$$

where D ranges over all effective $\mathbb{Q}$ -Weil divisors $\mathbb{Q}$ -linearly equivalent to Γ . When $\Delta = 0$, we write $\alpha_F(X; \Gamma)$ instead. Moreover, $\alpha_F(X; -K_X)$ is simply denoted by $\alpha_F(X)$.

Remark 3.20. We can define a positive characteristic analog of δ -invariants in a way similar to Definition 3.19. However, this definition differs from that given by Hattori–Odaka [HO24, Definition 2.16]. Their definition involves F -pure thresholds, whereas ours uses global F -split thresholds, which are in general strictly smaller than F -pure thresholds (see Remark 2.7).

We relate the F -alpha invariant for a projective variety to that for its section ring.

Proposition 3.21 (cf. [Pan23, Lemma 3.7]). *With notation as in Definition 3.19, assume that Γ is an ample $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -Weil divisor on X and set $S := \bigoplus_{i \geq 0} H^0(X, \mathcal{O}_X([i\Gamma]))$. Fix an integer $r \geq 1$ such that $r\Gamma$ is Cartier, and consider the r th Veronese subring $S^{(r)}$ of S . Let $\Delta_{S^{(r)}}$ denote the effective $\mathbb{Q}$ -Weil divisor on $\text{Spec } S^{(r)}$ corresponding to Δ . Then*

$$\alpha_F((X, \Delta); \Gamma) = r\alpha_F(S^{(r)}, \Delta_{S^{(r)}}).$$

Proof. For a real number $t > 0$, the condition $t \leq \alpha_F((X, \Delta); \Gamma)$ holds if and only if the pair $(X, \Delta + (t/(nr) - \varepsilon) \text{div } f)$ is globally F -regular for all integers $n \geq 1$, all sufficiently small real numbers $\varepsilon > 0$ and all nonzero elements $f \in H^0(X, \mathcal{O}_X(rn\Gamma))$. By [SS10, Proposition 5.3], the latter condition is equivalent to saying that the pair $(S^{(r)}, \Delta_{S^{(r)}} + (t/(nr) - \varepsilon) \text{div } f)$ is strongly F -regular for all such n, ε and f , which in turn holds if and only if $t \leq r\alpha_F(S^{(r)}, \Delta_S)$. $\square$

4. F -SIGNATURE IN EQUAL CHARACTERISTIC ZERO

In this section, we define the notion of F -signature in equal characteristic zero, and study its positivity using ultraproducts.

Definition 4.1. Let $(R, \mathfrak{m})$ be a local domain essentially of finite type over $\mathbb{C}$. Suppose that (A, R_A) is a model of R . Then F -signature $s(R)$ of R is defined as

$$s(R) = \sup_U \inf_{\mu \in U} s(R_\mu),$$

where U runs through all dense open subsets of $\text{Spec } A$ and μ runs through all closed points of U . This definition is independent of the choice of the model.

Proposition 4.2. *Definition 4.1 is independent of the choice of model.*

Proof. For a model $(A, R_A, \mathfrak{m}_A)$ of $(R, \mathfrak{m})$, we temporarily write

$$s_A(R) := \sup_U \inf_{\mu \in U} s(R_\mu).$$

Let $(B, R_B, \mathfrak{m}_B)$ be another model of $(R, \mathfrak{m})$. We will show that $s_A(R) = s_B(R)$. The proof is similar to an argument in [MS11, Remark 2.5].

First note that $s_A(R)$ remains unchanged after localizing A at a single element. Let $x \in X$ be a point of a complex algebraic variety X such that $\mathcal{O}_{X,x} \cong R$. Suppose that $(x_A \in X_A)$ is a model of $(x \in X)$ over A . Thus, localizing A at a single element, we may assume that $\mathfrak{m}_A(R_A \otimes_A A/\mu)$ is a prime ideal for all closed points $\mu \in \text{Spec } A$.

We can take a larger model $(C, R_C, \mathfrak{m}_C)$ satisfying the following conditions:

- (i) $A \subseteq C$,
- (ii) $R_C \cong R_A \otimes_A C$,
- (iii) $\mathfrak{m}_A R_C = \mathfrak{m}_C$,
- (iv) $\mathfrak{m}_C(R_C \otimes_C C/\nu)$ is a prime ideal for all closed points $\nu \in \text{Spec } C$.

It then suffices to show that $s_A(R) = s_C(R)$. Let $\varphi : \text{Spec } C \rightarrow \text{Spec } A$ be the morphism induced by the inclusion $A \hookrightarrow C$. Since C is of finite type over $\mathbb{Z}$, every

closed point $\nu \in \operatorname{Spec} C$ is sent by φ to a closed point $\mu := \varphi(\nu) \in \operatorname{Spec} A$. Therefore, we have a faithfully flat ring homomorphism

$$R_A \otimes_A A/\mu \rightarrow R_A \otimes_A C/\nu \cong R_C \otimes_C C/\nu,$$

which induces a flat local homomorphism

$$R_\mu := (R_A \otimes_A A/\mu)_{\mathfrak{m}_A(R_A \otimes_A A/\mu)} \rightarrow R_\nu := (R_C \otimes_C C/\nu)_{\mathfrak{m}_C(R_C \otimes_C C/\nu)}.$$

The closed fiber of the map $R_\mu \rightarrow R_\nu$ is a field, so it follows from [Yao06, Theorem 5.6] that $s(R_\mu) = s(R_\nu)$.

We now prove that $s_A(R) = s_C(R)$. The inequality $s_A(R) \leq s_C(R)$ can be checked easily. Indeed, for every dense open subset U of $\operatorname{Spec} A$, set $V := \varphi^{-1}(U)$. Since every closed point in V is sent to a closed point in U ,

$$\inf_{\mu \in U} s(R_\mu) \leq \inf_{\nu \in V} s(R_\nu),$$

where μ and ν run through all closed points of U and V , respectively. Therefore, $s_A(R) \leq s_C(R)$.

We show the converse inequality $s_A(R) \geq s_C(R)$. Let V be a dense open subset of $\operatorname{Spec} C$, and take a dense open subset U of $\operatorname{Spec} A$ such that $U \subseteq \varphi(V)$; such a U exists because φ is dominant. For every closed point μ , the fiber $V \cap \varphi^{-1}(\mu)$ contains a closed point ν of $\operatorname{Spec} C$. Hence,

$$\inf_{\mu \in U} s(R_\mu) \geq \inf_{\nu \in V} s(R_\nu),$$

where μ and ν run through all closed points of U and V , respectively. Thus, $s_A(R) \geq s_C(R)$. $\square$

Remark 4.3. Suppose that $R = (\mathbb{C}[x_1, \dots, x_n]/(f_1, \dots, f_r))_{(x_1, \dots, x_n)}$ is the localization at the maximal ideal $(x_1, \dots, x_n)$ of a quotient ring of a polynomial ring $\mathbb{C}[x_1, \dots, x_n]$ over $\mathbb{C}$ and that all coefficients of the defining equations f_i are integers. For each prime number p , let $R_p = (\mathbb{F}_p[x_1, \dots, x_n]/(\overline{f_1}, \dots, \overline{f_r}))_{(x_1, \dots, x_n)}$, where $\overline{f_i}$ is the image of f_i in the polynomial ring $\mathbb{F}_p[x_1, \dots, x_n]$ over $\mathbb{F}_p$. Then

$$s(R) = \liminf_{p \rightarrow \infty} s(R_p).$$

We record some basic properties of F -signature in equal characteristic zero.

Proposition 4.4. *Let $(R, \mathfrak{m})$ be a d -dimensional local domain essentially of finite type over $\mathbb{C}$.*

- (1) *If $s(R) > 0$, then R is of strongly F -regular type.*
- (2) *$s(R) = 1$ if and only if R is regular.*
- (3) *Suppose that R is normal $\mathbb{Q}$ -Gorenstein and $f \in \mathfrak{m}$ is a nonzero element. Then*

$$s(R) \geq s(R/(f)).$$

Proof. We use the same notation as in Definition 4.1.

- (1) This is immediate from Proposition 2.9 (3).

- (2) If R is regular, then there exists a dense open subset U of A such that R_μ is regular for every closed point $\mu \in U$. Therefore, by Proposition 2.9 (2), $s(R) = 1$.

For the converse, suppose that $s(R) = 1$ and R is not regular. Then by (1), there exists a dense open subset $U \subseteq \operatorname{Spec} A$ such that R_μ is a d -dimensional, F -finite, normal, Cohen–Macaulay, non-regular local domain for every closed point $\mu \in U$. It follows from [Tuc12, Theorem 4.14] (see also [HL02, Proposition 14]) that

$$s(R_\mu) \leq \frac{e(R_\mu) - e_{\text{HK}}(R_\mu)}{e(R_\mu) - 1}.$$

On the other hand, by [AE08, Theorem 4.12],

$$e_{\text{HK}}(R_\mu) \geq 1 + \frac{1}{d \cdot (d!(d-1) + 1)^d}.$$

This contradicts the assumption that $s(R) = 1$. Thus, R is regular.

- (3) Let $S := R/(f)$. By (1), we may assume that S is of strongly F -regular type. After enlarging A if necessary, we may choose a model S_A of S over A . By [Gro66, Théorème 12.2.4] and the argument in the proof of [HH20, Proposition 4.3.11(c)], there exists a dense open subset $U \subseteq \operatorname{Spec} A$ such that R_μ and S_μ are both normal and R_μ is $\mathbb{Q}$ -Gorenstein for all closed points $\mu \in U$. It then follows from [Tay20, Corollary 1.2] that $s(R_\mu) \geq s(S_\mu)$ for all closed points $\mu \in U$, which implies the assertion. $\square$

Rings of strongly F -regular type conjecturally coincide with singularities of klt type (see, for example, [HW02, 4.6]). Therefore, assuming this conjecture, we see from Proposition 4.4 (1) that if $s(R) > 0$, then $\operatorname{Spec} R$ is of klt type. Carvajal-Rojas, Schwede and Tucker ask in [CRST18, Question 5.10] whether the converse holds.

Conjecture A_d (cf. [CRST18, Question 5.10]). *Let $(R, \mathfrak{m})$ be a d -dimensional local domain essentially of finite type over $\mathbb{C}$. If $\operatorname{Spec} R$ is of klt type, then $s(R) > 0$.*

We simply say that Conjecture A holds if Conjecture A_d holds for all positive integers d .

Proposition 4.5. *Conjecture A_2 holds when R has residue field isomorphic to $\mathbb{C}$.*

Proof. First, note that two-dimensional singularities of klt type are $\mathbb{Q}$ -Gorenstein. Therefore, taking the index one cover of $\operatorname{Spec} R$, we may assume by [CRST18, Theorem 3.1] that R is Gorenstein, that is, $\operatorname{Spec} R$ is a rational double point.

Let $x \in X$ be a closed point of a complex algebraic variety X such that $\mathcal{O}_{X,x} \cong R$. Suppose that $(x_A \in X_A)$ is a model of $(x \in X)$ over a finitely generated $\mathbb{Z}$ -subalgebra A of $\mathbb{C}$. Let $(x_{\bar{\mu}} \in X_{\bar{\mu}})$ be the base change of $(x_\mu \in X_\mu)$ to an algebraic closure $\overline{\kappa(\mu)}$ of the base field $\kappa(\mu)$. Then, after enlarging A if necessary, we may assume that $(x_{\bar{\mu}} \in X_{\bar{\mu}})$ is a rational double point of the same type as $\operatorname{Spec} R$ for all closed points $\mu \in \operatorname{Spec} A$. Since the F -signature of $(x_{\bar{\mu}} \in X_{\bar{\mu}})$ depends only on its ADE type

and not on the characteristic of the residue field (if it is greater than 5) by [HL02, Example 18], we may further assume, after possibly enlarging A again, that there exists a constant $C > 0$ such that $s(\mathcal{O}_{X_{\bar{\mu}}, x_{\bar{\mu}}}) > C$ for all μ . By [Yao06, Theorem 5.6], this implies that $s(\mathcal{O}_{X_{\mu}, x_{\mu}}) > C$ for all μ , and therefore, $s(R) > C$. $\square$

Remark 4.6. It follows from [Sin05, Theorem 1.1] that Conjecture A holds for localizations of toric rings. This case is also covered by our main result Theorem 4.7.

The following is the main result of this section, which says that the uniform positivity of F -signature descends under pure morphisms.

Theorem 4.7. *Let $(R, \mathfrak{m}) \rightarrow (S, \mathfrak{n})$ be a pure local $\mathbb{C}$ -algebra homomorphism between normal local rings essentially of finite type over $\mathbb{C}$. If $\text{Spec } S$ is of klt type and $s(S) > 0$, then $s(R) > 0$.*

The proof of Theorem 4.7 is deferred to near the end of this section, following Proposition 4.15.

We apply Theorem 4.7 to the case of reductive quotient maps. A *reductive quotient singularity* over $\mathbb{C}$ is the spectrum $\text{Spec } R$ of a local $\mathbb{C}$ -algebra $(R, \mathfrak{m})$ such that its $\mathfrak{m}$ -adic completion $\widehat{R}$ is isomorphic to the ring of invariants $\mathbb{C}[[x_1, \dots, x_n]]^G$ where G is a linearly reductive group over $\mathbb{C}$ acting on a formal power series ring $\mathbb{C}[[x_1, \dots, x_n]]$.

The following fact is well known to experts, but we include it here due to the lack of a reference.

Lemma 4.8. *Let k be an algebraically closed field, S be a formal power series ring $k[[x_1, \dots, x_n]]$ over k and G be a linearly reductive group acting on the k -algebra S . Then the action of G on S can be linearized, i.e., there exist a k -algebra automorphism θ on S and a group inclusion $\sigma : G \hookrightarrow \text{GL}_n(k)$ such that the diagram*

$$\begin{array}{ccc} S & \xrightarrow{\sigma(g) \cdot} & S \\ \theta \downarrow & & \downarrow \theta \\ S & \xrightarrow{g \cdot} & S \end{array}$$

commutes for all $g \in G$, where $\text{GL}_n(k)$ acts naturally on S .

Proof. This follows from an argument similar to that in [Sat12, Proof of Corollary 1.8]. Let $\mathfrak{n}$ be the maximal ideal of S . Since G acts on $\mathfrak{n}/\mathfrak{n}^2$, we have an n -dimensional G -representation $\sigma : G \rightarrow \text{GL}_n(k)$. As G is linearly reductive, for every $i \geq 2$, there exists a G -equivariant k -linear section $s_i : \mathfrak{n}/\mathfrak{n}^i \rightarrow \mathfrak{n}/\mathfrak{n}^{i+1}$ of the natural surjection $\mathfrak{n}/\mathfrak{n}^{i+1} \twoheadrightarrow \mathfrak{n}/\mathfrak{n}^i$. These maps induce a G -equivariant k -linear section $s : \mathfrak{n}/\mathfrak{n}^2 \rightarrow \mathfrak{n}$ of the natural surjection $\mathfrak{n} \rightarrow \mathfrak{n}/\mathfrak{n}^2$. Define a k -algebra endomorphism $\theta : S \rightarrow S$ by $x_i \mapsto s(x_i + \mathfrak{n}^2)$ for each i . Then θ is an automorphism because $s(x_i + \mathfrak{n}^2) + \mathfrak{n}^2 = x_i + \mathfrak{n}^2$. $\square$

As an immediate corollary of Theorem 4.7, we prove that Conjecture A holds for reductive quotient singularities over $\mathbb{C}$.

Corollary 4.9. *Let $(R, \mathfrak{m})$ be a local domain essentially of finite type over $\mathbb{C}$. If $\text{Spec } R$ is a reductive quotient singularity over $\mathbb{C}$, then $s(R) > 0$.*

Proof. We use the following fact:

Claim. Let $(R, \mathfrak{m})$ be a local domain essentially of finite type over $\mathbb{C}$. If its $\mathfrak{m}$ -adic completion $\widehat{R}$ is a pure subring of a regular local ring $(B, \mathfrak{m}_B)$, then R itself is a pure subring of some regular local ring essentially of finite type over $\mathbb{C}$.

Proof of Claim. It follows from Popescu's theorem that B can be written as a direct limit $\varinjlim_{\lambda \in \Lambda} B_\lambda$ of smooth $\mathbb{C}$ -algebras B_λ . Let A be a domain of finite type over $\mathbb{C}$ and $\mathfrak{p} \subset A$ be a prime ideal such that $R \cong A_{\mathfrak{p}}$. Then the natural injection $A \rightarrow B$ factors through B_{λ_0} for some $\lambda_0 \in \Lambda$ (see [SPA, Tag 07C3]). Letting S be the localization of B_{λ_0} at the prime ideal $\mathfrak{m}_B \cap B_{\lambda_0}$, we see that $R \rightarrow S$ is a pure local homomorphism. $\square$

Suppose that R is a reductive quotient singularity over $\mathbb{C}$. By Lemma 4.8, the completion $\widehat{R}$ of R is a pure subring of a formal power series ring $\mathbb{C}[[x_1, \dots, x_n]]$. Applying the above claim to our R , we see that R is a pure subring of a regular local ring essentially of finite type over $\mathbb{C}$. Therefore, it follows from Proposition 4.4 and Theorem 4.7 that $s(R) > 0$. $\square$

In order to show lemmas necessary for the proof of Theorem 4.7, we work with the following setting.

Situation 4.10. Let $(R, \mathfrak{m}) \hookrightarrow (S, \mathfrak{n})$ be a pure local $\mathbb{C}$ -algebra homomorphism between normal local rings essentially of finite type over $\mathbb{C}$, and let d denote the dimension of R . Fix a non-principal ultrafilter $\mathcal{F}$ on the set $\mathcal{P}$ of prime numbers and a field isomorphism $\gamma: \text{ulim}_p \overline{\mathbb{F}_p} \xrightarrow{\sim} \mathbb{C}$. Let $(R_p)_{p \in \mathcal{P}}$ and $(S_p)_{p \in \mathcal{P}}$ be approximations of R and S , respectively.

As a key ingredient in the proof of Theorem 4.7, we introduce a generalization of ultra- F -regularity, originally introduced by Schoutens [Sch05].

Definition 4.11. With notation as in Situation 4.10, let f be a nonzero element of R_∞ and $t_p > 0$ be a rational number for almost all p , and set $t := \text{ulim}_p t_p$. We then say that the pair (R, f^t) is *ultra- F -regular* if for every nonzero element $c \in R_\infty$, there exists a non-standard natural number $\varepsilon \in {}^*\mathbb{N}$ such that the R -module homomorphism

$$R \rightarrow F_*^\varepsilon R_\infty; \quad x \longmapsto F_*^\varepsilon (cf^{\lceil t\pi^\varepsilon \rceil} F^\varepsilon(x))$$

is pure.

Remark 4.12. With notation as in Definition 4.11, suppose that f is a nonzero element of R and almost all t_p are equal to some rational number t_0 . Then t can be identified with t_0 , and (R, f^t) is ultra- F -regular in the sense of Definition 4.11 if and only if the pair (R, f^{t_0}) is ultra- F -regular in the sense of [Yam23, Definition 5.5].

The following proposition explains the relationship between the ultra- F -regularity of a local ring essentially of finite type over $\mathbb{C}$ and the strong F -regularity of its approximations.

Proposition 4.13 (cf. [Yam23, Proposition 5.5]). *With notation as in Situation 4.10, let f_p be a nonzero element of R_p and t_p be a positive rational number for almost all p , and set $f = \text{ulim}_p f_p$ and $t = \text{ulim}_p t_p$.*

- (1) *If the pair $(R_p, t_p \text{div}(f_p))$ is strongly F -regular for almost all p , then (R, f^t) is ultra- F -regular.*
- (2) *Suppose that the anti-canonical ring $\mathcal{R} := \bigoplus_{i \geq 0} R(-iK_X)$ of $X := \text{Spec } R$ is finitely generated. Then the pair $(R_p, t_p \text{div}(f_p))$ is strongly F -regular for almost all p if and only if (R, f^t) is ultra- F -regular.*

Proof. (1) Let η be a socle generator for $H_{\mathfrak{m}}^d(\omega_R)$ and $(\eta_p)_{p \in \mathcal{P}}$ be its approximation. Here note that a generator of the socle is unique up to multiplication by a unit. Then η_p is a socle generator for $H_{\mathfrak{m}_p}^d(\omega_{R_p})$ for almost all p , because the natural map $H_{\mathfrak{m}}^d(\omega_R) \rightarrow \text{ulim}_p H_{\mathfrak{m}_p}^d(\omega_{R_p})$ is injective (see Propositions 3.3, 3.5 in [Yam25a] and the proof of Proposition 3.9 in loc. cit.). Fix a nonzero element $c = \text{ulim}_p c_p$ of R_{∞} . For almost all p , the pair $(R_p, t_p \text{div}(f_p))$ is strongly F -regular, so there exists an integer $e_p \geq 1$ such that $\eta_p \otimes F_*^{e_p}(c_p f_p^{\lceil t p^{e_p} \rceil}) \neq 0$ in $H_{\mathfrak{m}_p}^d(\omega_{R_p}) \otimes_{R_p} F_*^{e_p} R_p$. Letting $\varepsilon := \text{ulim}_p e_p$, we have the following commutative diagram:

$$\begin{array}{ccc} H_{\mathfrak{m}}^d(\omega_R) & \longrightarrow & H_{\mathfrak{m}}^d(\omega_R) \otimes_R F_*^{\varepsilon} R_{\infty} \\ \downarrow & & \downarrow \\ \text{ulim}_p H_{\mathfrak{m}_p}^d(\omega_{R_p}) & \longrightarrow & \text{ulim}_p (H_{\mathfrak{m}_p}^d(\omega_{R_p}) \otimes_{R_p} F_*^{e_p} R_p). \end{array}$$

Here the top horizontal map is given by $\xi \mapsto \xi \otimes F_*^{\varepsilon}(c f^{\lceil t \pi^{\varepsilon} \rceil})$ and the bottom horizontal map is given by $\text{ulim}_p \xi_p \mapsto \text{ulim}_p (\xi_p \otimes F_*^{e_p}(c_p f_p^{\lceil t p^{e_p} \rceil}))$. Under the right vertical map, we have

$$\eta \otimes F_*^{\varepsilon}(c f^{\lceil t \pi^{\varepsilon} \rceil}) \mapsto \text{ulim}_p (\eta_p \otimes F_*^{e_p}(c_p f_p^{\lceil t p^{e_p} \rceil})).$$

Since the image is nonzero, the element $\eta \otimes F_*^{\varepsilon}(c f^{\lceil t \pi^{\varepsilon} \rceil})$ is nonzero in $H_{\mathfrak{m}}^d(\omega_R) \otimes_R F_*^{\varepsilon} R_{\infty}$. It follows that the map $R \rightarrow F_*^{\varepsilon} R_{\infty}$ sending 1 to $F_*^{\varepsilon}(c f^{\lceil t \pi^{\varepsilon} \rceil})$ is pure.

(2) We prove the “if” part. We may assume that $-K_X$ is an effective Weil divisor on X , so that $R(-iK_X)$ is an ideal of R for all $i \geq 0$. Let $\mathcal{M} := \mathfrak{m} + \mathcal{R}_{>0}$ be the unique homogeneous maximal ideal of $\mathcal{R}$. First, we show the following claim.

Claim. If (R, f^t) is ultra- F -regular, then so is $(\mathcal{R}_{\mathcal{M}}, f^t)$.

Proof of Claim. We identify $H_{\mathcal{M}}^{d+1}(\omega_{\mathcal{R}})$ with $\bigoplus_{i \geq 0} H_{\mathfrak{m}}^d(R(iK_X))$ by [Wat94, Theorem 2.2]. Note also that $\text{Soc}_R H_{\mathfrak{m}}^d(\omega_R) = \text{Soc}_{\mathcal{R}} H_{\mathcal{M}}^{d+1}(\omega_{\mathcal{R}})$. Let $c \in \mathcal{R}$ be a nonzero element, and fix any integer $i \geq 0$ such that the i -th homogeneous part c_i of c is nonzero. Since (R, f^t) is ultra- F -regular, there exists a non-standard natural number $\varepsilon \in {}^*\mathbb{N}$ such that the R -linear map

$$c_i f^{\lceil t \pi^{\varepsilon} \rceil} F^{\varepsilon} : R \rightarrow F_*^{\varepsilon} R_{\infty}; \quad 1 \mapsto F_*^{\varepsilon}(c_i f^{\lceil t \pi^{\varepsilon} \rceil})$$

is pure. On the other hand, it follows from [Yam25b, Proposition 3.8] that there exists an R -linear map

$$\psi : H_{\mathcal{M}}^{d+1}(\omega_{\mathcal{R}}) \otimes_{\mathcal{R}} F_*^{\varepsilon}(\mathcal{R}_{\mathcal{M}})_{\infty} \rightarrow H_{\mathfrak{m}}^d(\omega_R) \otimes_R F_*^{\varepsilon} R_{\infty}$$

sending $\xi \otimes F_*^\varepsilon(cg)$ to $\xi \otimes F_*^\varepsilon(c_i g)$ for each $\xi \in H_{\mathfrak{m}}^d(\omega_R) \subseteq H_{\mathcal{M}}^{d+1}(\omega_{\mathcal{R}})$ and $g \in R_\infty$. Therefore, we have the following commutative diagram:

$$\begin{array}{ccc} H_{\mathcal{M}}^{d+1}(\omega_{\mathcal{R}}) & \longrightarrow & H_{\mathcal{M}}^{d+1}(\omega_{\mathcal{R}}) \otimes_{\mathcal{R}} F_*^\varepsilon(\mathcal{R}_{\mathcal{M}})_\infty \\ \downarrow & & \downarrow \psi \\ H_{\mathfrak{m}}^d(\omega_R) & \xrightarrow{1 \otimes c_i f^{[t\pi^\varepsilon]} F^\varepsilon} & H_{\mathfrak{m}}^d(\omega_R) \otimes_R F_*^\varepsilon R_\infty, \end{array}$$

where the top horizontal map sends ξ to $\xi \otimes F_*^\varepsilon(c f^{[t\pi^\varepsilon]})$ and the bottom horizontal map sends η to $\eta \otimes F_*^\varepsilon(c_i f^{[t\pi^\varepsilon]})$. The bottom map is injective by the purity of $c_i f^{[t\pi^\varepsilon]} F^\varepsilon$, and the socle of $H_{\mathcal{M}}^{d+1}(\omega_{\mathcal{R}})$ coincides with that of $H_{\mathfrak{m}}^d(\omega_R)$, so the top map is also injective. Thus, $(\mathcal{R}_{\mathcal{M}}, f^t)$ is ultra- F -regular.

Suppose that (R, f^t) is ultra- F -regular. By the above claim, $(\mathcal{R}_{\mathcal{M}}, f^t)$ is also ultra- F -regular. Suppose that $((\mathcal{R}_{\mathcal{M}})_p, t_p \operatorname{div}(f_p))$ is not strongly F -regular for almost all p . Then, for almost all p , there exists a nonzero element $c_p \in \mathcal{R}_p$ such that for any $e \geq 0$, $c_p f_p^{[tp^e]} F^e : (\mathcal{R}_{\mathcal{M}})_p \rightarrow F_*^e(\mathcal{R}_{\mathcal{M}})_p$ is not pure. Let $c := \operatorname{ulim}_p c_p$. Since $\mathcal{R}$ is quasi-Gorenstein by [GHN90, Theorem 4.5], we have

$$H_{\mathcal{M}}^{d+1}(\omega_{\mathcal{R}}) \otimes_{\mathcal{R}} M \cong H_{\mathcal{M}}^{d+1}(\mathcal{R}) \otimes_{\mathcal{R}} M \cong H_{\mathcal{M}}^{d+1}(M)$$

for every $\mathcal{R}$ -module M , where the second isomorphism follows from the fact the $(\dim \mathcal{R})$ -th local cohomology module is computed as the rightmost cohomology of the Čech complex associated to $\mathcal{R}$. Thus, for each $\varepsilon \in {}^*\mathbb{N}$, there exists the following commutative diagram:

$$\begin{array}{ccccc} H_{\mathcal{M}}^{d+1}(\omega_{\mathcal{R}}) & \xrightarrow{1 \otimes c f^{[t\pi^\varepsilon]} F^\varepsilon} & H_{\mathcal{M}}^{d+1}(\omega_{\mathcal{R}}) \otimes_{\mathcal{R}} F_*^\varepsilon(\mathcal{R}_{\mathcal{M}})_\infty & \xrightarrow{\cong} & H_{\mathcal{M}}^{d+1}(F_*^\varepsilon(\mathcal{R}_{\mathcal{M}})_\infty) \\ \downarrow & & \downarrow & & \downarrow \\ \operatorname{ulim}_p H_{\mathcal{M}_p}^{d+1}(\omega_{\mathcal{R}_p}) & \rightarrow & \operatorname{ulim}_p (H_{\mathcal{M}_p}^{d+1}(\omega_{\mathcal{R}_p}) \otimes_{\mathcal{R}_p} F_*^{e_p}(\mathcal{R}_{\mathcal{M}})_p) & \xrightarrow{\cong} & \operatorname{ulim}_p H_{\mathcal{M}_p}^{d+1}(F_*^{e_p}(\mathcal{R}_{\mathcal{M}})_p). \end{array}$$

Since the right vertical map is injective by [Yam25a, Proposition 3.9], $1 \otimes c f^{[t\pi^\varepsilon]} F^\varepsilon$ is not injective, a contradiction. Hence, $((\mathcal{R}_{\mathcal{M}})_p, t_p \operatorname{div}(f_p))$ is strongly F -regular for almost all p . For almost all p , the ring $(\mathcal{R}_{\mathcal{M}})_p$ is the localization at the unique homogeneous maximal ideal of an $\mathbb{N}$ -graded R_p -algebra whose degree-zero component is isomorphic to R_p , so the natural homomorphism $R_p \rightarrow (\mathcal{R}_{\mathcal{M}})_p$ is pure. Therefore, $(R_p, t_p \operatorname{div}(f_p))$ is strongly F -regular for almost all p . $\square$

The authors learned the proof of the following lemma from Kenta Sato, who they thank.

Lemma 4.14. *With notation as in Theorem 4.7, suppose that $(R_A, \mathfrak{m}_A, S_A, \mathfrak{n}_A)$ is a model of $(R, \mathfrak{m}, S, \mathfrak{n})$ over a finitely generated $\mathbb{Z}$ -subalgebra $A \subseteq \mathbb{C}$. Then there exists a constant $C > 0$ and a dense open subset $U \subseteq \operatorname{Spec} A$ such that*

$$C \cdot \overline{\operatorname{ord}}_{\mathfrak{m}_\mu}(f) \geq \overline{\operatorname{ord}}_{\mathfrak{n}_\mu}(f)$$

for all closed points $\mu \in U$ and all nonzero elements $f \in R_\mu$.

Proof. By [Hüb01, Corollary 2], there exists a divisorial valuation v of the fractional field $\text{Frac}(R)$ of R , centered on $\mathfrak{m}$, that extends to a divisorial valuation w of the fractional field $\text{Frac}(S)$ of S , centered on $\mathfrak{n}$. Enlarging A if necessary, we may take a model (v_A, w_A) of (v, w) over A . Since w_μ is a divisorial valuation centered on $\mathfrak{n}_\mu$ for general closed points $\mu \in \text{Spec } A$, we have $w_\mu(\overline{\mathfrak{n}_\mu^n}) \geq n$ for every integer $n \geq 1$. In particular,

$$v_\mu(f) = w_\mu(f) \geq \overline{\text{ord}}_{\mathfrak{n}_\mu}(f)$$

for all nonzero elements $f \in R_\mu$ and such μ .

Let $\pi : Y \rightarrow X = \text{Spec } R$ be a log resolution of $\mathfrak{m}$ with exceptional divisor $E = \sum_{i=1}^r E_i$ such that v corresponds to E_1 and $\mathfrak{m}\mathcal{O}_Y = \mathcal{O}_Y(-\sum_{i=1}^r a_i E_i)$. We also fix a π -ample Cartier divisor H on Y . Enlarging A again, we may take a model $(\pi_A : Y_A \rightarrow X_A, E_A = \sum_{i=1}^r E_{i,A}, H_A)$ of $(\pi : Y \rightarrow X, E = \sum_{i=1}^r E_i, H)$ over A . Then for general closed points $\mu \in \text{Spec } A$, the morphism $\pi_\mu : Y_\mu \rightarrow X_\mu = \text{Spec } R_\mu$ is a log resolution of $\mathfrak{m}_\mu$ with exceptional divisor $E_\mu = \sum_{i=1}^r E_{i,\mu}$ such that v_μ corresponds to $E_{1,\mu}$ and $\mathfrak{m}_\mu\mathcal{O}_{Y,\mu} = \mathcal{O}_{Y,\mu}(-\sum_{i=1}^r a_i E_{i,\mu})$. Moreover, H_μ is a π_μ -ample Cartier divisor on Y_μ .

We now apply Izumi's theorem. Using essentially the same argument as in the proof of Izumi's theorem given in [BFJ14]¹, there exists a constant $C_\mu > 0$ such that

$$C_\mu \cdot \overline{\text{ord}}_{\mathfrak{m}_\mu}(f) \geq v_\mu(f)$$

for all nonzero elements $f \in R_\mu$. Note that C_μ is determined by the intersection numbers $(H_\mu^{d-2} \cdot E_{i,\mu} \cdot E_{j,\mu})$, as well as a_1 and r . Since these intersection numbers agree with $(H^{d-2} \cdot E_i \cdot E_j)$ for general closed points $\mu \in \text{Spec } A$, we can choose the same constant C (independent of μ) for such μ . This completes the proof. $\square$

Proposition 4.15. *With notation as in Situation 4.10, suppose that the anti-canonical ring $\mathcal{R} = \bigoplus_{i \geq 0} R(-iK_X)$ of $X := \text{Spec } R$ is finitely generated. Then there exists a constant $C > 0$ such that $C \cdot \alpha_F(R_p) \geq \alpha_F(S_p)$ for almost all p .*

Proof. Choose $C > 0$ to be a constant satisfying the inequality in Lemma 4.14. By Remark 2.26, together with the fact that normalized orders do not change under faithfully flat extensions (see, for example, [HS06, Proposition 1.6.2]), we have

$$C \cdot \overline{\text{ord}}_{\mathfrak{m}_p}(f) \geq \overline{\text{ord}}_{\mathfrak{n}_p}(f)$$

for almost all p .

Suppose to the contrary that for almost all p , the inequality $C \cdot \alpha_F(R_p) < \alpha_F(S_p)$ holds. Then, for each such p , there exists a nonzero element $f_p \in \mathfrak{m}_p$ such that

$$C \cdot \text{fpt}_{R_p}(f_p) \overline{\text{ord}}_{\mathfrak{m}_p}(f_p) < \alpha_F(S_p) \leq \text{fpt}_{S_p}(f_p) \overline{\text{ord}}_{\mathfrak{n}_p}(f_p).$$

By the choice of C , we deduce that $\text{fpt}_{R_p}(f_p) < \text{fpt}_{S_p}(f_p)$ for almost all p . Choose a rational number t_p lying in the interval $(\text{fpt}_{R_p}(f_p), \text{fpt}_{S_p}(f_p))$. Then the pair

¹The base field is assumed to have characteristic zero in *loc. cit.*, but this is only used to guarantee the existence of log resolutions. Since we already have a log resolution π_μ , the argument applies in our setting as well.

$(S_p, t_p \operatorname{div}(f_p))$ is strongly F -regular for almost all p , which is equivalent, by Proposition 4.13, to saying that (S, f^t) is ultra- F -regular, where $f = \operatorname{ulim}_p f_p$ and $t = \operatorname{ulim}_p t_p$. Therefore, for every nonzero element $c \in R \subseteq S$, there exists a non-standard natural number $\varepsilon \in {}^*\mathbb{N}$ such that the S -linear map $S \rightarrow F_*^\varepsilon S_\infty$ sending x to $F_*^\varepsilon(c f^{\lceil t p^\varepsilon \rceil} F^\varepsilon(x))$ is pure. Now we consider the following commutative diagram:

$$\begin{array}{ccc} R & \longrightarrow & F_*^\varepsilon R_\infty \\ \downarrow & & \downarrow \\ S & \longrightarrow & F_*^\varepsilon S_\infty, \end{array}$$

where the horizontal maps send x to $F_*^\varepsilon(c f^{\lceil t p^\varepsilon \rceil} F^\varepsilon(x))$. Since the inclusion map $R \rightarrow S$ is pure, the top horizontal map is also pure, which means that (R, f^t) is ultra- F -regular. Applying Proposition 4.13 again, we conclude that $(R_p, t_p \operatorname{div}(f_p))$ is strongly F -regular for almost all p . However, this contradicts the choice of t_p . $\square$

We are now ready to prove the main result of this section, Theorem 4.7.

Proof of Theorem 4.7. First note that $\operatorname{Spec} R$ is also of klt type by a result of Zhuang [Zhu24, Theorem 1.1]. Hence, the anti-canonical ring of R is finitely generated by [BCHM10]. Suppose that $(R_A, \mathfrak{m}_A)$ is a model of $(R, \mathfrak{m})$ over a finitely generated $\mathbb{Z}$ -subalgebra $A \subseteq \mathbb{C}$. Assume to the contrary that $s(R) = 0$, that is, for every constant $C > 0$, there exists a dense subset U_C of closed points of $\operatorname{Spec} A$ such that $s(R_\mu) \leq C$ for all $\mu \in U_C$. Then, by an argument similar to the proof of [Yam25a, Proposition 4.6], we can construct a dense subset $U = \{\mu_i\}_{i=1}^\infty$ of closed points of $\operatorname{Spec} A$, a non-principal ultrafilter $\mathcal{F}$ on $\mathcal{P}$ and a field isomorphism $\gamma : \operatorname{ulim}_p \overline{\mathbb{F}}_p \xrightarrow{\sim} \mathbb{C}$ satisfying the following 4 conditions:

- (i) $p_i < p_{i+1}$ for all $i \geq 1$,
- (ii) $s(R_{\mu_i}) \leq 1/i$ for all $i \geq 1$,
- (iii) $\{p_i\}_{i=1}^\infty \in \mathcal{F}$,
- (iv) a natural map $R_{\mu_i} \rightarrow R_{p_i}$ is flat local and $\mathfrak{m}_{\mu_i} R_{p_i} = \mathfrak{m}_{p_i}$,

where p_i is the characteristic of the residue field A/μ_i and $((R_p, \mathfrak{m}_p))_{p \in \mathcal{P}}$ is an approximation of the local ring $(R, \mathfrak{m})$ with respect to $\mathcal{F}$ and γ .

Claim. Let $((S_p, \mathfrak{n}_p))_{p \in \mathcal{P}}$ be an approximation of $(S, \mathfrak{n})$ with respect to $\mathcal{F}$ and γ . Then there exists a constant $C_0 > 0$ such that $s(S_p) > C_0$ for almost all p .

Proof of Claim. After enlarging A if necessary, choose a model $(S_A, \mathfrak{n}_A)$ of $(S, \mathfrak{n})$ over A . Fix a constant C_0 satisfying $0 < C_0 < s(S)$. Then there exists a dense open subset $V \subseteq \operatorname{Spec} A$ such that $s(S_\mu) > C_0$ for all closed points $\mu \in V$. For almost all p , by Remark 2.26, one can find a closed point $\mu_p \in V$ and a flat local homomorphism $S_{\mu_p} \rightarrow S_p$ with $\mathfrak{m}_{\mu_p} S_p = \mathfrak{n}_p$. It then follows from [Yao06, Theorem 5.6] that $s(S_p) = s(S_{\mu_p}) > C_0$ for almost all p . $\square$

Combining the above claim with Proposition 3.9, we find a constant $C_1 > 0$ such that $\alpha_F(S_p) > C_1$ for almost all p . Note that N in Proposition 3.9 (2) depends only on the constant appearing in Izumi's theorem, which, by the proof of Lemma 4.14,

does not depend on p . Then Proposition 4.15 yields a constant $C_2 > 0$ such that $\alpha_F(R_p) > C_2$ for almost all p . Applying Proposition 3.9, we obtain a constant $C_3 > 0$ such that $s(R_p) > C_3$ for almost all p . Finally, by [Yao06, Theorem 5.6] again, we have

$$0 < C_3 \leq s(R_{p_i}) = s(R_{\mu_i}) < \frac{1}{i}$$

for infinitely many $i \geq 1$, which is a contradiction. $\square$

We close this section with the following corollary concerning the Gorenstein case of Conjecture A. We say that Conjecture B holds if Conjecture A holds for Gorenstein rings.

Conjecture B_d. *Let $(R, \mathfrak{m})$ be a d -dimensional Gorenstein local domain essentially of finite type over $\mathbb{C}$. If $\text{Spec } R$ has klt singularities, then $s(R) > 0$.*

Corollary 4.16. *For each integer $d \geq 2$, Conjecture B_{d+1} implies Conjecture A_d.*

Proof. Let R be as in Conjecture A_d. Let $\mathcal{R} := \bigoplus_{i \geq 0} R(-iK_X)$ denote the anti-canonical ring of $X := \text{Spec } R$ and $\mathcal{M}$ denote the unique homogeneous maximal ideal of $\mathcal{R}$. Since X is of klt type, it follows from a result of [BCHM10] that $\mathcal{R}$ is a finitely generated R -algebra and $\text{Spec } \mathcal{R}_{\mathcal{M}}$ is a Gorenstein klt singularity of dimension $d+1$. Therefore, by Conjecture B_{d+1}, we have $s(\mathcal{R}_{\mathcal{M}}) > 0$. Noting that the natural map $R \rightarrow \mathcal{R}_{\mathcal{M}}$ is a split local homomorphism, we see from Theorem 4.7 that $s(R) > 0$. $\square$

5. LOCAL TO GLOBAL

In this section, we discuss a global interpretation of Conjecture A introduced in the previous section.

First, we introduce the notion of F -alpha invariants in equal characteristic zero, in a way similar to Definition 4.1, which defines F -signature in equal characteristic zero.

Definition 5.1. Let X be a projective variety over $\mathbb{C}$, Δ be an effective $\mathbb{Q}$ -Weil divisor on X and Γ be a big $\mathbb{Q}$ -Weil divisor on X . Suppose that $(X_A, \Delta_A, \Gamma_A)$ is a model of (X, Δ, Γ) over a finitely generated $\mathbb{Z}$ -subalgebra A of $\mathbb{C}$. Then F -alpha invariant $\alpha_F((X, \Delta); \Gamma)$ is defined as

$$\alpha_F((X, \Delta); \Gamma) = \sup_U \inf_{\mu \in U} \alpha_F((X_\mu, \Delta_\mu); \Gamma_\mu),$$

where U runs through all dense open subsets of $\text{Spec } A$ and μ runs through all closed points of U .

Conjecture C_d. *Let (X, Δ) be a d -dimensional projective log Fano pair over $\mathbb{C}$. Then*

$$\alpha_F((X, \Delta); -(K_X + \Delta)) > 0.$$

We simply say that Conjecture C holds if Conjecture C_d holds for all positive integers d .

Theorem 5.2. *Conjecture C implies Conjecture B. Also, for each integer $d \geq 2$, Conjecture C_{d-1} implies Conjecture B_d for local rings R with residue field $\mathbb{C}$.*

In order to prove Theorem 5.2, we work with the following setting.

Situation 5.3. Let x be a closed point of an affine Gorenstein klt variety X over $\mathbb{C}$. Let $\sigma : Y \rightarrow X$ be a projective birational morphism that provides a Kollár component F , that is, σ is an isomorphism over $X \setminus \{x\}$ and $\sigma^{-1}(\{x\})$ is a prime divisor F on Y such that the pair (Y, F) is plt and $-F$ is $\mathbb{Q}$ -Cartier σ -ample. For each integer $j \geq 0$, we define the ideal $\mathfrak{a}_j \subseteq \mathcal{O}_X$ as

$$\mathfrak{a}_j = \{f \in \mathcal{O}_X \mid \text{ord}_F(f) \geq j\} = H^0(Y, \mathcal{O}_Y(-jF)) \subseteq \mathcal{O}_X,$$

and let G denote the $\mathbb{N}$ -graded $\mathbb{C}$ -algebra $\bigoplus_{j \geq 0} \mathfrak{a}_j / \mathfrak{a}_{j+1}$. Since $-F$ is σ -ample, G is finitely generated.

Proposition 5.4. *With notation as in Situation 5.3, suppose that $(x_A \in X_A, G_A)$ is a model of $(x \in X, G)$ over a finitely generated $\mathbb{Z}$ -subalgebra $A \subseteq \mathbb{C}$. Then there exists a constant $C > 0$ such that*

$$\alpha_F(\mathcal{O}_{X_\mu, x_\mu}) \geq C \cdot \alpha_F(G_\mu)$$

for general closed points $\mu \in \text{Spec } A$.

Proof. We define the $\mathbb{Z}$ -graded $\mathcal{O}_X$ -algebra $\mathcal{R}'$ by

$$\mathcal{R}' = \bigoplus_{j \in \mathbb{Z}} \mathfrak{a}_j t^{-j} \subseteq \mathcal{O}_X[t, t^{-1}],$$

where $\mathfrak{a}_j = H^0(Y, \mathcal{O}_Y(-jF)) \subseteq \mathcal{O}_X$. Since F is σ -anti-ample, $\mathcal{R}'$ is finitely generated. The inclusion map $\mathbb{C}[t] \rightarrow \mathcal{R}'$ is flat, and we have the following isomorphisms.

$$\begin{aligned} \mathcal{R}' \otimes_{\mathbb{C}[t]} \mathbb{C}[t]/(t) &\cong G, \\ \mathcal{R}' \otimes_{\mathbb{C}[t]} \mathbb{C}[t, t^{-1}] &\cong \mathcal{O}_X[t, t^{-1}]. \end{aligned}$$

It follows from [LX20, Section 2.4] (see also [LZ22, Proposition 2.10]) that G is isomorphic, as an $\mathbb{N}$ -graded $\mathbb{C}$ -algebra, to $\bigoplus_{j \geq 0} H^0(F, \mathcal{O}_F([-jF]_F))$. Moreover, by [LZ22, Proposition 2.9], the spectrum of this graded ring has klt singularities.

Now, after enlarging A if necessary, we may choose a model $\mathcal{R}'_A$ of $\mathcal{R}'$ over A . We may further assume that for every closed point $\mu \in \text{Spec } A$, the ring $\mathcal{R}'_\mu$ is a finitely generated $\mathbb{Z}$ -graded $\mathcal{O}_{X_\mu}$ -subalgebra of $\mathcal{O}_{X_\mu}[t, t^{-1}]$ ², and the inclusion map $\kappa(\mu)[t] \rightarrow \mathcal{R}'_\mu$ is flat. For each integer $j \in \mathbb{Z}$, let $\text{pr}_j : \mathcal{O}_{X_\mu}[t, t^{-1}] \rightarrow \mathcal{O}_{X_\mu}$ denote the natural projection onto the degree j component, and define $\mathfrak{a}_{j,\mu}$ to be the ideal of $\mathcal{O}_{X_\mu}$ making the following commutative diagram commute:

$$\begin{array}{ccc} \mathcal{R}'_\mu & \hookrightarrow & \mathcal{O}_{X_\mu}[t, t^{-1}] \\ \downarrow \text{pr}_j & & \downarrow \text{pr}_j \\ \mathfrak{a}_{j,\mu} t^{-j} & \hookrightarrow & \mathcal{O}_{X_\mu} t^{-j}. \end{array}$$

²The fact that $\mathcal{R}'_\mu$ is a subalgebra of $\mathcal{O}_{X_\mu}[t, t^{-1}]$ follows from [HH20, Theorem 2.3.5(b)].

In particular, $\mathcal{R}'_\mu = \bigoplus_{j \in \mathbb{Z}} \mathfrak{a}_{j,\mu} t^{-j}$ and $G_\mu = \bigoplus_{j \geq 0} \mathfrak{a}_{j,\mu} / \mathfrak{a}_{j+1,\mu}$. Let

$$T_\mu := \mathcal{R}'_\mu \otimes_{\mathcal{O}_{X_\mu}} \mathcal{O}_{X_\mu, x_\mu} = \bigoplus_{j \in \mathbb{Z}} \mathfrak{a}_{j,\mu} \mathcal{O}_{X_\mu, x_\mu} t^{-j} \subseteq \mathcal{O}_{X_\mu, x_\mu} [t, t^{-1}].$$

Then T_μ is a $*$ local $\mathbb{Z}$ -graded $\mathcal{O}_{X_\mu, x_\mu}$ -algebra. Note that for general closed points $\mu \in \text{Spec } A$, we have the following isomorphisms:

$$\begin{aligned} T_\mu \otimes_{\kappa(\mu)[t]} \kappa(\mu)[t]/(t) &\cong (\mathcal{R}'_\mu \otimes_{\kappa(\mu)[t]} \kappa(\mu)[t]/(t)) \otimes_{\mathcal{O}_{X_\mu}} \mathcal{O}_{X_\mu, x_\mu} \\ &\cong (\mathcal{R}' \otimes_{\mathbb{C}[t]} \mathbb{C}[t]/(t))_\mu \otimes_{\mathcal{O}_{X_\mu}} \mathcal{O}_{X_\mu, x_\mu} \\ &\cong G_{\mu, x_\mu}, \\ T_\mu \otimes_{\kappa(\mu)[t]} \kappa(\mu)[t, t^{-1}] &\cong (\mathcal{R}'_\mu \otimes_{\kappa(\mu)[t]} \kappa(\mu)[t, t^{-1}]) \otimes_{\mathcal{O}_{X_\mu}} \mathcal{O}_{X_\mu, x_\mu} \\ &\cong (\mathcal{R}' \otimes_{\mathbb{C}[t]} \mathbb{C}[t, t^{-1}])_\mu \otimes_{\mathcal{O}_{X_\mu}} \mathcal{O}_{X_\mu, x_\mu} \\ &\cong \mathcal{O}_{X_\mu, x_\mu} [t, t^{-1}]. \end{aligned}$$

Since $\text{Spec}(\mathcal{R}' \otimes_{\mathbb{C}[t]} \mathbb{C}[t]/(t))$ has klt singularities, we deduce from Theorem 2.11 that there exists a dense open subset $U \subseteq \text{Spec } A$ such that G_{μ, x_μ} is strongly F -regular for all closed points $\mu \in U$.

Fix a closed point $\mu \in U$. For a nonzero element $f \in \mathcal{O}_{X_\mu, x_\mu}$, set

$$\begin{aligned} \tilde{f} &:= t^{-n} f \in \mathfrak{a}_{n,\mu} \mathcal{O}_{X_\mu, x_\mu} t^{-n} \subseteq T_\mu, \\ \text{in}(f) &:= [f] \in \mathfrak{a}_{n,\mu} \mathcal{O}_{X_\mu, x_\mu} / \mathfrak{a}_{n+1,\mu} \mathcal{O}_{X_\mu, x_\mu} \subseteq G_{\mu, x_\mu}, \end{aligned}$$

where n is the largest integer j with $f \in \mathfrak{a}_{j,\mu} \mathcal{O}_{X_\mu, x_\mu}$. Let $s > 0$ be a real number such that the pair $(G_{\mu, x_\mu}, \text{in}(f)^s)$ is strongly F -regular, and in particular, F -rational. It then follows from a combination of [ST08, Propositions 6.15(1), 7.3] and Lemma 2.14 that $(T_\mu, \tilde{f}^s)$ is also F -rational. Since $\mathcal{O}_{X_\mu, x_\mu} [t, t^{-1}]$ is a localization of T_μ , the pair $(\mathcal{O}_{X_\mu, x_\mu} [t, t^{-1}], f^s)$ is likewise F -rational. By Remark 2.13 and the fact that $\mathcal{O}_{X_\mu, x_\mu} [t, t^{-1}]$ is Gorenstein, this implies that $(\mathcal{O}_{X_\mu, x_\mu} [t, t^{-1}], f^s)$ is strongly F -regular. Finally, using the faithful flatness of the natural map $\mathcal{O}_{X_\mu, x_\mu} \rightarrow \mathcal{O}_{X_\mu, x_\mu} [t, t^{-1}]$, we conclude that $(\mathcal{O}_{X_\mu, x_\mu}, f^s)$ is strongly F -regular as well. Putting everything together, we obtain the inequality

$$\text{fpt}(\mathcal{O}_{X_\mu, x_\mu}, f) \geq \text{fpt}(G_{\mu, x_\mu}, \text{in}(f)).$$

Meanwhile, let $\mathfrak{m}_x$ be the maximal ideal of $\mathcal{O}_X$ corresponding to the closed point x . By the uniform version of Izumi's theorem (see the proof of Lemma 4.14), there exists a constant $C > 0$ such that for general closed points $\mu \in \text{Spec } A$, we have $\mathfrak{a}_{n,\mu} \subseteq \mathfrak{m}_{x_\mu}^{[Cn]}$ for all integers $n \geq 1$. As a consequence, every nonzero element $f \in \mathcal{O}_{X_\mu, x_\mu}$ satisfies that $\text{ord}_{\mathfrak{m}_{x_\mu}}(f) \geq C \cdot \deg(\text{in}(f))$, which leads to the inequality

$$\text{fpt}(\mathcal{O}_{X_\mu, x_\mu}, f) \text{ord}_{\mathfrak{m}_{x_\mu}}(f) \geq C \cdot \text{fpt}(G_{\mu, x_\mu}, \text{in}(f)) \deg(\text{in}(f)).$$

Thus, $\alpha_F(\mathcal{O}_{X_\mu, x_\mu}) \geq C \cdot \alpha_F(G_\mu)$ for general closed points $\mu \in \text{Spec } A$. $\square$

Proposition 5.5. *With notation as in Situation 5.3, let $a \geq 0$ be an integer such that $K_Y = \sigma^* K_X + aF$ and Δ_F denote the $\mathbb{Q}$ -Weil divisor on F known as*

the different on F (see [Kol13, Definition 2.34] for its definition). Suppose that $(\sigma_A: Y_A \rightarrow X_A, F_A, \Delta_{F,A}, G_A)$ is a model of $(\sigma: Y \rightarrow X, F, \Delta_F, G)$ over a finitely generated $\mathbb{Z}$ -subalgebra $A \subseteq \mathbb{C}$. Then

$$\alpha_F(G_\mu) = (1+a) \cdot \alpha_F((F_\mu, \Delta_{F,\mu}); -(K_{F_\mu} + \Delta_{F,\mu}))$$

for general closed points $\mu \in \text{Spec } A$.

Proof. Given a $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -Weil divisor B on a normal variety Z , the Cartier index of B at a point $z \in Z$ is denoted by $\text{ind}(z \in Z, B)$. Let y be a codimension one point of F and D be the prime divisor on F corresponding to y . It follows from [HLS26, Theorem A.1] that

$$\text{ind}(y \in Y, F) = \text{ind}(y \in F, F|_F).$$

Therefore, if we set $m := \text{ind}(y \in Y, F)$, then there exists an integer $n \geq 1$ such that $\text{ord}_D(F|_F) = n/m$ and $\gcd(m, n) = 1$. Applying this argument together with [K⁺92, Proposition 16.6] we have prime divisors D_i on F , positive integers m_i and integers n_i such that

$$\Delta_F = \sum_i \frac{m_i - 1}{m_i} D_i, \quad F|_F = \sum_i \frac{n_i}{m_i} D_i,$$

and $\gcd(m_i, n_i) = 1$ for each i .

We consider the following two graded rings:

$$S := \bigoplus_{i \geq 0} H^0(F, \mathcal{O}_F(-ir(K_F + \Delta_F))),$$

$$S' := \bigoplus_{i \geq 0} H^0(F, \mathcal{O}_F(\lfloor -iF|_F \rfloor)),$$

where $r = \text{ind}(K_F + \Delta_F)$. By [LZ22, Proposition 2.10], S' is isomorphic to G as a graded $\mathbb{C}$ -algebra. Also, by the definition of Δ_F ,

$$K_F + \Delta_F = (K_Y + F)|_F = (\sigma^* K_X + (1+a)F)|_F = (1+a)F|_F.$$

Therefore, S is a Veronese subring of S' . Let $\pi: \text{Spec } S' \rightarrow \text{Spec } S$ denote the morphism induced by the inclusion and Δ_S denote the $\mathbb{Q}$ -Weil divisor on S corresponding to Δ_F . It follows from [Wat81, (1.3) and Theorem 2.8] that $K_{S'} = (\pi')^*(K_S + \Delta_S)$.

Now we reduce the setting modulo p . We may assume that the degree of the finite map $\pi_\mu: \text{Spec } S'_\mu \rightarrow \text{Spec } S_\mu$ is not divisible by the characteristic of $\kappa(\mu)$, so that the trace map $\text{Tr}_{\pi_\mu}: \text{Frac}(S'_\mu) \rightarrow \text{Frac}(S_\mu)$ is surjective, where $\text{Frac}(S'_\mu)$ (resp. $\text{Frac}(S_\mu)$) denotes the quotient field of S'_μ (resp. S_μ). Since

$$\alpha_F((F_\mu, \Delta_{F,\mu}); -(K_{F_\mu} + \Delta_{F,\mu})) = r \cdot \alpha_F(S_\mu, \Delta_{S_\mu})$$

by Proposition 3.21, it suffices to prove that

$$\alpha_F(G_\mu) = \alpha_F(S'_\mu) = (1+a)r \cdot \alpha_F(S_\mu, \Delta_{S_\mu})$$

for general closed points $\mu \in \text{Spec } A$. To prove this, it is enough to show that $\text{fpt}(S'_\mu, f) = \text{fpt}((S_\mu, \Delta_{S_\mu}); f)$ for each nonzero nonunit homogeneous element $f \in S_\mu$.

For every real number $t \geq 0$, [ST14, Main Theorem] gives the equality

$$\mathrm{Tr}_{\pi_\mu}(\tau(S'_\mu, f^t)) = \tau((S_\mu, \Delta_{S_\mu}); f^t).$$

By the surjectivity of the trace map Tr_{π_μ} , we have $\mathrm{fpt}(S'_\mu, f) = \mathrm{fpt}((S_\mu, \Delta_{S_\mu}); f)$. $\square$

Proof of Theorem 5.2. Let $(R, \mathfrak{m})$ be a Gorenstein local domain essentially of finite type over $\mathbb{C}$. Take a point x on a Gorenstein algebraic variety X over $\mathbb{C}$ such that $\mathcal{O}_{X,x} \cong R$. By replacing x with a closed point of the closure $\overline{\{x\}}$, we may assume that x is a closed point of X . We now use the same notation as in Situation 5.3. Then the assertion follows immediately from Propositions 5.4 and 5.5, since

$$\alpha_F(R) \geq C \cdot \alpha_F(G) \geq C(1+a) \cdot \alpha_F((F, \Delta_F); -(K_F + \Delta_F)),$$

where $C > 0$ is the constant from Proposition 5.4 and $a \geq 0$ is the integer such that $K_Y = \sigma^* K_X + aF$. $\square$

In the latter half of this section, we consider two weaker versions of Conjecture C_d : the first concerns varieties of Fano type, and the second concerns $\mathbb{Q}$ -Fano varieties.

Conjecture D_d . *If X is a d -dimensional complex variety of Fano type, then $\alpha_F(X) > 0$.*

Conjecture E_d . *If X is a d -dimensional $\mathbb{Q}$ -Fano variety over $\mathbb{C}$, then $\alpha_F(X) > 0$.*

We investigate the relationship between the above conjectures and those discussed in the previous sections.

Proposition 5.6. (1) *Conjecture B_{d+1} for local rings R with residue field $\mathbb{C}$ implies that Conjecture E_d .*

(2) *Conjecture C_d implies Conjecture D_d .*

Proof. (1) Let X be a d -dimensional $\mathbb{Q}$ -Fano variety over $\mathbb{C}$. By Lemma 3.15 and Proposition 3.21, applying Conjecture B_{d+1} to the localization of the anti-canonical ring $R(X, -K_X)$ of X at its homogeneous maximal ideal yields $\alpha_F(X) > 0$.

(2) Let X be a complex projective variety of Fano type. It suffices to show that if $\alpha_F(X; -(K_X + \Delta)) > 0$, then $\alpha_F(X) > 0$. Since $-(n-1)K_X - n\Delta = -n(K_X + \Delta) - (-K_X)$ is ample for sufficiently large integers n , by Lemma 5.7, we obtain

$$\alpha_F(X) \geq \alpha_F(X; -n(K_X + \Delta)) = \frac{1}{n} \alpha_F(X; -(K_X + \Delta)) > 0.$$

$\square$

Lemma 5.7. *Let X be a globally F -regular projective variety over an F -finite field of characteristic $p > 0$, Γ be a big $\mathbb{Q}$ -Weil divisor on X and E be a $\mathbb{Q}$ -effective $\mathbb{Q}$ -Weil divisor on X . Then*

$$\alpha_F(X; \Gamma) \geq \alpha_F(X; \Gamma + E).$$

Proof. Let t be a real number greater than $\alpha_F(X; \Gamma)$. Then there exist effective $\mathbb{Q}$ -Weil divisors Δ and D on X such that $\Delta \sim_{\mathbb{Q}} E$, $D \sim_{\mathbb{Q}} \Gamma$ and $t \geq \mathrm{gfst}(X, D)$. Then $D + \Delta \sim_{\mathbb{Q}} \Gamma + E$ and $t \geq \mathrm{gfst}(X, D) \geq \mathrm{gfst}(X, D + \Delta)$, which implies that $t \geq \alpha_F(X; \Gamma + E)$. Therefore, $\alpha_F(X; \Gamma) \geq \alpha_F(X; \Gamma + E)$. $\square$

The following theorem is the main result of the latter half of this section.

Theorem 5.8. *Conjecture D_d is equivalent to Conjecture E_d .*

To prove Theorem 5.8, we first establish a couple of lemmas.

Lemma 5.9. *Suppose that $\pi : X \dashrightarrow Y$ is a small birational map between globally F -regular projective varieties over an F -finite field of characteristic $p > 0$ and Γ is a big $\mathbb{Q}$ -Weil divisor on X . Then*

$$\alpha_F(X; \Gamma) = \alpha_F(Y; \pi_* \Gamma).$$

Proof. This follows from an argument similar to [GOST15, Lemma 2.12]. $\square$

Lemma 5.10. *Let $\pi : X \rightarrow Y$ be a birational morphism between $\mathbb{Q}$ -Gorenstein globally F -regular projective varieties over an F -finite field of characteristic $p > 0$. Suppose that the exceptional locus E of π is a prime divisor and $K_X - \pi^* K_Y \leq 0$. Then*

$$\alpha_F(X) \geq \min\{\alpha_F(Y), 1\}.$$

Proof. Let D be an effective $\mathbb{Q}$ -Cartier $\mathbb{Q}$ -Weil divisor on X that is $\mathbb{Q}$ -linearly equivalent to $-K_X$. Choose an integer $n \geq 1$ such that nD is Cartier and nD is linearly equivalent to $-nK_X$. It then suffices to show that $\text{gfst}(X, D) \geq t$ for every rational number $t \in (0, \min\{\alpha_F(Y), 1\}) \cap n\mathbb{Z}_{(p)}$. By assumption,

$$\pi^* K_Y = K_X + aE,$$

where a is a nonnegative rational number. Set $D' = \pi_* D$. Then $D' \sim_{\mathbb{Q}} -K_Y$, so D' is $\mathbb{Q}$ -Cartier and we have $\pi^* D' = D - aE$.

Let C be any effective Cartier divisor on X , and take an effective Cartier divisor C' on Y such that $\pi^* C' \geq C$. Since $t < \alpha_F(Y)$ and $t \in n\mathbb{Z}_{(p)}$, there exists an integer $e \geq 1$ such that $(p^e - 1)t/n \in \mathbb{Z}$ and the map

$$\mathcal{O}_Y \rightarrow F_*^e \mathcal{O}_Y((p^e - 1)tD' + C')$$

splits as an $\mathcal{O}_Y$ -module homomorphism. The splitting homomorphism φ lies in

$$\begin{aligned} & \text{Hom}_{\mathcal{O}_Y}(F_*^e \mathcal{O}_Y((p^e - 1)tD' + C'), \mathcal{O}_Y) \\ & \cong H^0(Y, F_*^e \mathcal{O}_Y((1 - p^e)K_Y - (p^e - 1)tD' - C')) \\ & = H^0(Y, \pi_* F_*^e \mathcal{O}_X(\pi^*((1 - p^e)K_Y - (p^e - 1)tD' - C'))) \\ & \subseteq H^0(X, F_*^e \mathcal{O}_X((1 - p^e)K_X - (p^e - 1)tD - C)) \\ & \cong \text{Hom}_{\mathcal{O}_X}(F_*^e \mathcal{O}_X((p^e - 1)tD + C), \mathcal{O}_X). \end{aligned}$$

Here we use the fact that for any Weil divisor G on an F -finite normal variety Z , the adjunction formula (cf. the proof of [HW02, Theorem 3.3]) gives

$$\mathcal{H}om_{\mathcal{O}_Z}(F_*^e \mathcal{O}_Z(G), \mathcal{O}_Z) \cong F_*^e \mathcal{O}_Z((1 - p^e)K_Z - G).$$

Therefore, φ also gives a splitting of the map $\mathcal{O}_X \rightarrow F_*^e \mathcal{O}_X((p^e - 1)tD + C)$, which completes the proof. $\square$

Proof of Theorem 5.8. It is clear that Conjecture D_d implies Conjecture E_d , so we will prove the converse implication. Let X be a d -dimensional complex projective variety of Fano type, that is, there exists an effective $\mathbb{Q}$ -Weil divisor Δ on X such that (X, Δ) is klt and $-(K_X + \Delta)$ is ample. By Lemma 5.9, after taking a $\mathbb{Q}$ -factorization, we may assume that X is $\mathbb{Q}$ -factorial. Since X is a Mori dream space by [BCHM10], we can run a $(-K_X)$ -minimal model program and obtain a sequence of divisorial contractions and flips:

$$X = X_0 \xrightarrow{f_0} X_1 \xrightarrow{f_1} \cdots \xrightarrow{f_{l-2}} X_{l-1} \xrightarrow{f_{l-1}} X_l = X',$$

where each X_i is of Fano type and $-K_{X'}$ is semi-ample. $-K_{X'}$ induces a birational contraction $g : X' \rightarrow Z$ such that $g^*K_Z = K_{X'}$ and $Z = \text{Proj } R(X, -K_X)$ is the anti-canonical model of X .

Now we reduce the situation to characteristic $p > 0$. Suppose that $(X_A, f_{j,A} : X_{j,A} \dashrightarrow X_{j+1,A}, Z_A)$ is a model of $(X, f_j : X_j \dashrightarrow X_{j+1}, Z)$ over a finitely generated $\mathbb{Z}$ -subalgebra $A \subseteq \mathbb{C}$. If f_j is a divisorial contraction, then K_{X_j} is f_j -ample, so the Negativity Lemma implies that $f_j^*K_{X_{j+1}} - K_{X_j}$ is effective. Applying Lemma 5.10, we then have $\alpha_F(X_{j,\mu}) \geq \min\{\alpha_F(X_{j+1,\mu}), 1\}$ for general closed points $\mu \in \text{Spec } A$. If f_j is a flip, then by Lemma 5.9, we have $\alpha_F(X_{j,\mu}) = \alpha_F(X_{j+1,\mu})$ for general closed points $\mu \in \text{Spec } A$. Therefore,

$$\alpha_F(X_\mu) \geq \min\{\alpha_F(X'_\mu), 1\} \geq \min\{\alpha_F(Z_\mu), 1\}$$

for general closed points $\mu \in \text{Spec } A$, where the second inequality again follows from Lemma 5.10. Finally, by applying Conjecture E_d to Z , we conclude that $\alpha_F(X) > 0$. $\square$

Summing up, we have seen that the following implications hold among the conjectures discussed in this paper:

$$\begin{array}{ccccc} & & \text{Conjecture } A_{d+1} & & \\ & & \Downarrow & & \\ \text{Conjecture } C_d & \xrightarrow{5.2^3} & \text{Conjecture } B_{d+1} & \xrightarrow{4.16} & \text{Conjecture } A_d \\ \Downarrow 5.6(2) & & \Downarrow 5.6(1) & & \\ \text{Conjecture } D_d & \xleftrightarrow{5.8} & \text{Conjecture } E_d & & \end{array}$$

We conclude this section with the following question.

Question 5.11. Does Conjecture D_d imply Conjecture C_d ? If so, Conjecture A would be equivalent to Conjecture E and could be reduced to the case where R is a Gorenstein graded ring.

Remark 5.12. A key difficulty in the above question is that the global F -split region

$$\text{cl}(\{(s, t) \in \mathbb{R}_{\geq 0}^2 \mid (V, sB + tD) \text{ is globally } F\text{-regular}\}),$$

³More precisely, this implication holds when the ring has residue field isomorphic to $\mathbb{C}$.

where $\text{cl}(-)$ denotes closure in the Euclidean topology, V is a globally F -regular variety and B, D are effective $\mathbb{Q}$ -Weil divisors on V , is not a polytope unlike log canonical regions.

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